Response to Reviewer 1

Draft date: 2026-10-19

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We thank Reviewer 1 for the careful and technically detailed assessment. The comments identified several places where the original manuscript and released code did not sufficiently document the provenance, comparability, or reproducibility of the reported results. We have treated these comments as an opportunity to audit the underlying calculations, clarify the scope of the claims, and strengthen both the manuscript and the code release.

This remains a working response letter. Values already verified in the Revision 1 audit are reported explicitly. Items that require additional cluster calculations, retraining, or final integration remain indicated by bracketed placeholders; their calculation routes are tracked in `pending_values_register.csv`.

Overall response to Reviewer 1's summary assessment

We agree with the reviewer that this study should be presented as a task-conditioned case study rather than as evidence for a universal ordering of specialist and foundation MLIPs. We have therefore reframed the benchmark as a hierarchical evaluation ladder, together with complementary non-equilibrium probes, rather than as a single scalar ranking of model classes.

The first level is a static single-point baseline: energy and force RMSEs are evaluated on DFT test configurations and provide the conventional pointwise-accuracy reference (Table S1 and Fig. S1). These metrics establish local interpolation accuracy, but do not by themselves test trajectory or transition-state fidelity.

The second level consists of equilibrium and near-equilibrium molecular-dynamics tests. We compare RDFs, temperature and pressure stability, VACF, MSD, and related structural and thermodynamic observables at 600 K (Fig. 2 and Fig. S2). The four MACE models reproduce the principal local coordination features. The zero-shot foundation model nevertheless shows a persistent pressure offset, and transport estimates are not pooled because the trajectory lengths and sampling protocols are not fully matched.

The third level is fixed-path NEB evaluation. DFT first defines a relaxed migration pathway, and each MLIP is evaluated on the same fixed intermediate geometries. We distinguish an interpolative within-gap probe from higher-distortion candidates that move toward or into a covalently bonded quintuple layer (QL); the latter are extrapolative probes and are not pooled into a final ranking while their reference-quality audits remain open. On the controlled within-gap probe, the grouped three-seed absolute fixed-path barrier errors are 0.47 ± 0.12 eV for FT-600K, 1.20 ± 0.27 eV for Scratch, and 0.68 ± 0.20 eV for the Scratch-5% control. The historical 0.16 eV

point estimate is not reproduced by the grouped multi-seed evaluation.

The fourth level is self-relaxed NEB. In this protocol, the MLIP is not given the DFT-relaxed intermediate images and must generate its own pathway through model-in-the-loop force feedback. We now distinguish this operator explicitly from fixed-path energy evaluation. The historical SI Fig. S3 record is retained only as a provenance audit because it corresponds to a different system and image sequence and contains calculator and final-band serialization defects. It is therefore excluded from quantitative model ranking.

As a complementary nonlocal out-of-distribution probe, we evaluate collective interlayer sliding (Fig. 5). This test shows that fine-tuning that improves local migration chemistry does not necessarily preserve long-range registry physics. A matched fixed-path screen of ten foundation checkpoints further shows target-specific barrier miscalibration: four representative checkpoints predict barriers of 1.50, 1.79, 1.60, and 1.84 eV, compared with the current provisional DFT candidate of 0.34 eV. Their absolute errors range from 1.17 to 1.50 eV.

Taken together, the results support a conditional decision framework rather than a universal model ranking. A foundation prior can improve data efficiency after task-relevant adaptation for moderate distribution shifts, but it can remain target-miscalibrated and can lose long-range response. Specialist from-scratch models can provide target-specific calibration but may exhibit stability risks outside their training support. Higher-distortion pathways remain an unresolved extrapolation test because converged DFT references and path-level seed statistics are incomplete. These scope limitations are now stated explicitly in the title, abstract, Introduction, Figure 1, Results, and Supplementary Information.

A1. Fig. 3d versus Fig. S3: NEB barrier discrepancy

Reviewer comment.

> *"Reconcile the foundation-model in-gap barrier discrepancy: state explicitly whether Fig. 3d (fixed-geometry, ∼0.7 eV overestimate, 'unacceptable') and Fig. S3 (self-run NEB, 0.41 eV, 'exceptional') use the same pathway and DFT reference; fix the reference value (0.34 vs. ∼0.3 eV); present both evaluations side-by-side and explain why relaxation changes the verdict. As written, the paper's central ranking is ambiguous."*

Response. We agree that the original manuscript did not distinguish the fixed-path and self-relaxed NEB evaluations clearly enough. In the revised benchmark hierarchy, fixed-path NEB is the third-level process probe, whereas self-relaxed NEB is a more demanding fourth-level probe. Our audit also identified a provenance problem in the historical SI Fig. S3 record, so the two records are not treated as directly comparable quantitative barriers.

- **Fixed-path NEB evaluation (main-text Fig. 3d).** The DFT-relaxed NEB images are held fixed, and each MLIP is used only to evaluate the energies of those same geometries. This tests energetic calibration on a common DFT-defined migration path. The audited path is the 61-atom, one-Cr system $CrSb_{24}Te_{36}$ with five DFT-relaxed images. The primary case is a within-gap, interpolative probe; higher-distortion gap-to-QL/into-QL candidates are treated as extrapolative probes.

$$\Delta E_M^{\mathrm{fixed}}(\Gamma_{\mathrm{DFT}}) = \max_i E_M(X_i^{\mathrm{DFT}}) - E_M(X_0^{\mathrm{DFT}}). \tag{1}$$

- **Self-relaxed NEB evaluation (historical SI Fig. S3).** In a valid self-relaxed comparison, the MLIP would start from the endpoint structures and relax its own elastic band using its own forces and energy landscape:

$$\Gamma_M^\star = \mathrm{NEB}_M(X_0, X_1), \qquad \Delta E_M^{\mathrm{self}} = \max_i E_M(X_i^{M,\star}) - E_M(X_0). \tag{2}$$

Thus, Fig. 3d evaluates energetic calibration on a fixed DFT path, whereas a valid Fig. S3-style test evaluates model-in-the-loop path generation. The historical Fig. S3 record cannot be used for this comparison because its system and calculator lineage do not match the main-text path.

For the fixed-path within-gap probe, the archived QE `neb.out` calculation gives a DFT barrier of 0.34 eV at manuscript precision. The archived NEB calculation is numerically stationary but does not yet meet our final convergence criterion, so this value is reported as a provisional pathway-specific candidate. The full-precision value is retained in the raw audit files. A final restartable QE closure calculation will either confirm or replace this value and all derived signed errors.

Using the current audited within-gap reference, the fixed-geometry results are:

Table 1: Audited fixed-geometry results for the 61-atom within-gap probe.

Model	Barrier (eV)	DFT candidate (eV)	Signed error (eV)
DFT candidate	0.34	0.34	0.00
Foundation	1.02	0.34	+0.69
Scratch	4.54	0.34	+4.20
FT-600K	0.50	0.34	+0.16
FT-MultiT	0.82	0.34	+0.49

The DFT value of 0.34 eV is the current archived QE candidate and is reported as provisional because the final restartable QE convergence gate is not yet closed; the full-precision value is retained in the audit files. The historical 0.16 eV value is retained only as a frame-level historical result and is not the current grouped estimate. In the grouped, leakage-controlled re-evaluation, the three-seed absolute errors are 0.47 ± 0.12 eV for OMAT-FT, 1.20 ± 0.27 eV for Scratch, and 0.68 ± 0.20 eV for Scratch-5%.

The historical SI Fig. S3 record reported a Foundation value of 0.41 eV, but the provenance audit shows that it corresponds to the different system $Cr_2Sb_{32}Te_{48}$ with 82 atoms, two Cr dopants, and twelve images, rather than the 61-atom, one-Cr system used in Fig. 3d. The historical calculation also used inconsistent endpoint and internal-image calculators, and the final band contains a serialization defect. Uniform rescoring of the preserved band gives a forward endpoint-referenced value of 0.00 eV, a reverse value of 0.38 eV, and a whole-band span of 1.04 eV, rather than 0.41 eV. We therefore mark the historical value `EXCLUDE_FROM_PUBLICATION`; it is retained only as a provenance audit and is not used as evidence of exceptional generalization or as a valid self-relaxed barrier.

We do not claim that relaxation changes the verdict for the same physical path on the basis of the historical Fig. S3 record. The valid conceptual distinction remains important: fixed-path scoring tests energy assignment on DFT-relaxed images, whereas a valid self-consistent NEB test would assess model-in-the-loop path generation. A same-system self-consistent comparison,

including DFT rescoring of the MLIP-relaxed images, remains an explicit validation gap rather than a completed ranking result.

As an additional matched fixed-path check, we evaluated ten Foundation checkpoints (MP-0b3, MPA0, OMAT-0 small/medium, MH-0/MH-1, Agnesi medium, and three Agnesi stress/density variants) on the same five within-gap images. The four representative checkpoints MP-0b3, MPA0, OMAT-0 medium, and MH-1 gave barriers of 1.50, 1.79, 1.60, and 1.84 eV, respectively, relative to the provisional 0.34 eV DFT candidate. Their absolute errors therefore range from 1.17 to 1.50 eV, and MPA0 is not the kinetic-accuracy winner in this completed comparison. The ten individual profiles and family-aggregated profiles are added to the SI. The family bands represent sub-model variation, not seed uncertainty; all foundation checkpoints are single-checkpoint comparisons. The complete preflight manifest, raw profiles, profile diagnostics, and SHA256 ledger are released with the revision artifacts.

We have added a side-by-side protocol and provenance audit with the pathway/system identity, atom and image counts, image source, relaxation and energy engines, reported barrier, DFT reference, and scientific status. The audit figure is included in the revision evidence package as `fig3d_figS3_protocol_side_by_side_audited.pdf`; its right panel shows why the historical 0.41 eV number is excluded rather than treated as a competing barrier.

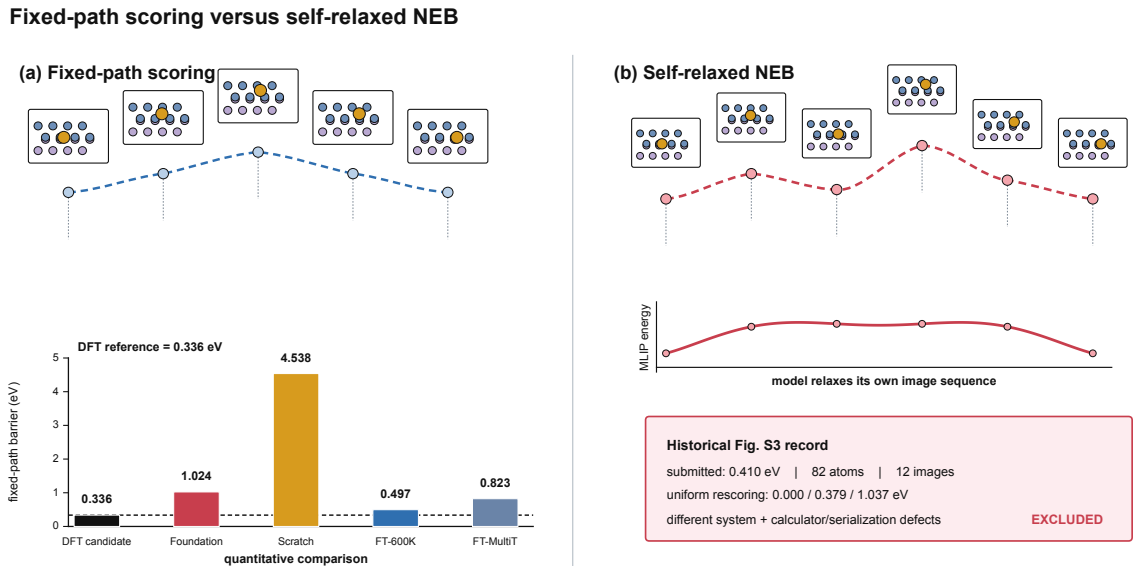
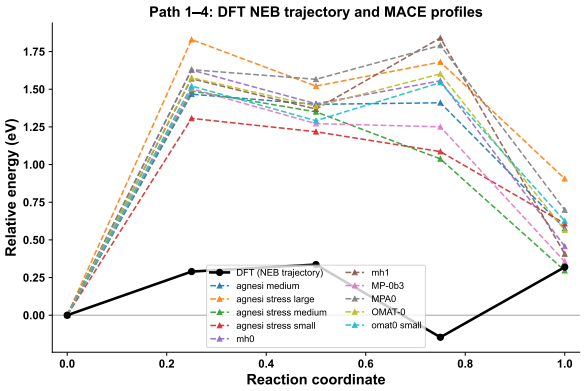


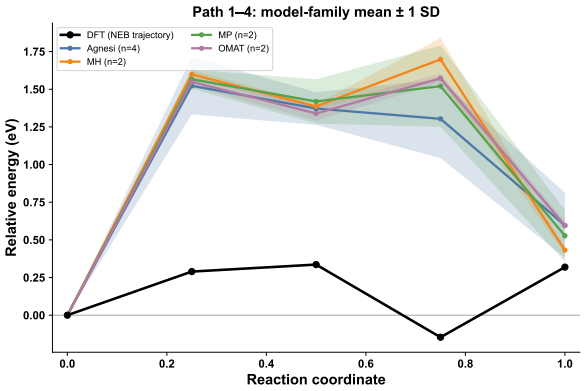
Figure 1: Fixed-path scoring versus self-relaxed NEB. (a) Fixed-path evaluation for the 61-atom, one-Cr within-gap probe. The DFT-relaxed images are held fixed, and the bars show the barriers assigned by the DFT reference candidate and the four MLIP strategies. The dashed line marks the provisional DFT reference of 0.34 eV. (b) Schematic of the self-relaxed NEB operator, in which intermediate images move under MLIP forces and generate a model-dependent energy profile. The audit box reports the historical SI Fig. S3 record: submitted value 0.41 eV, 82 atoms, twelve images, and uniform-rescoring values of 0.00, 0.38, and 1.04 eV. Because the historical record uses a different system and contains calculator and serialization defects, it is excluded from quantitative model ranking.

Reused review-book context for A1. The following historical assets are retained because they show the fixed-path profiles and the local PES/convergence diagnostics that motivated the audit. They are not additional self-relaxed barriers and do not replace the corrected side-by-side audit above.

Manuscript/SI change.

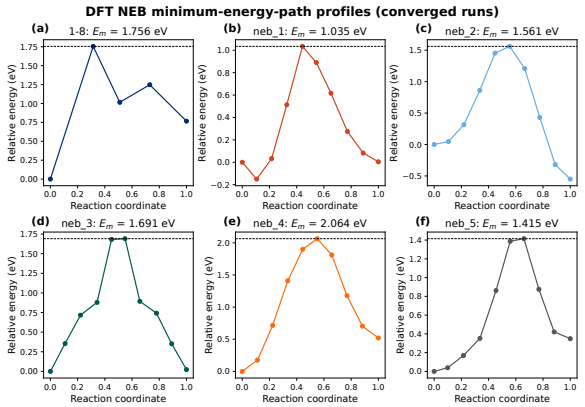


(a) FIG-194: corrected path-1–4 profiles.

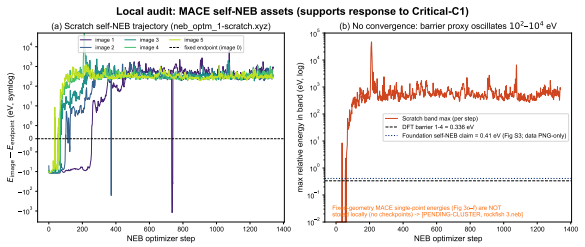


(b) FIG-195: family-aggregated profiles.

Figure 2: Reused fixed-path foundation assets. These review-book pages show the ten-checkpoint foundation comparison on the common DFT-defined path and the corresponding within-family spread. The panels support the fixed-path/operator distinction and the target-specific miscalibration discussion; they are single-checkpoint comparisons or family summaries, not training-seed uncertainty and not evidence for a valid self-relaxed NEB barrier. The current DFT reference remains provisional.



(a) FIG-153: DFT MEP profiles.



(b) FIG-155: local PES audit.

Figure 3: Reused DFT/PES audit assets. These existing SI assets document the local path-energy and PES diagnostics associated with the historical record. They are included for provenance and convergence context only; they do not repair the historical system mismatch, endpoint/internal-calculator mismatch, or final-band serialization defect, and they do not establish a same-system self-relaxed result.

Unchanged context: the fixed-path versus self-relaxed operator distinction remains central, while the historical SI record is retained only for provenance.

Inserted/revised manuscript text.

The historical SI Fig. S3 record reported a 0.41 eV Foundation value, but the Revision 1 audit found that it does not correspond to the main-text 61-atom within-gap system: it uses an 82-atom, two-Cr system with twelve images rather than the 61-atom, one-Cr, five-image path used for the fixed-geometry comparison. The historical calculation also used inconsistent endpoint and movable-image calculators, and the final band contains a serialization defect. Uniform rescoring of the preserved band gives a forward endpoint-referenced barrier of 0.00 eV, a reverse value of 0.38 eV, and a whole-band span of 1.04 eV, not 0.41 eV. We therefore mark the historical 0.41 eV value `EXCLUDE_FROM_PUBLICATION` and retain it only to document the source of the original discrepancy. A valid same-system self-consistent MLIP-NEB comparison remains a future validation target.

The response letter now places the corrected fixed-path table, the side-by-side audit, and the reused FIG-153/FIG-155/FIG-194/FIG-195 provenance pages immediately below this comment.

A2. Train/test overlap for FT-600K

Reviewer comment.

> *"Rule out train/test overlap for FT-600K: report the minimum SOAP-distance (or RMSD) between each in-gap NEB image and its nearest fine-tuning frame, or re-run FT-600K with the NEB-adjacent trajectory segment excluded, and confirm the 0.16 eV result survives."*

Response. We agree that the original frame-level shuffle did not rule out trajectory-adjacent overlap. We therefore rebuilt the split by source-trajectory provenance and re-evaluated the declared fine-tuning files. Every source trajectory is assigned in full to train, validation, or test, giving 1,786/311/267 configurations for the full grouped split. The Scratch-5% control uses the same validation and test partitions and an 89-configuration grouped training subset. The audit covers 25 fixed NEB images spanning the controlled within-gap probe and the available higher-distortion candidate classes; the internal audit identifiers are retained only in the SI and machine-readable evidence.

The overlap audit found zero exact global matches, zero exact Cr-local matches, and zero Cr-local RMSD values at or below 0.05 Å in both training splits. The minimum training-set SOAP cosine distances were 5.4×10^{-4} for the full grouped split and 6.4×10^{-4} for Scratch-5%; the minimum comparable Cr-local RMSDs were 2.06 Å for both splits. These values are above the declared near-duplicate thresholds (SOAP $\leq 10^{-4}$ and RMSD ≤ 0.05 Å). SOAP was evaluated for all target images; the RMSD comparison was restricted to species-composition-compatible local environments, with two comparable training candidates for the full split and one for Scratch-5%. We therefore describe the result as no detected overlap under the declared exact, SOAP, and RMSD metrics, rather than as a claim about the unavailable pretraining corpus.

We then retrained and evaluated FT-600K from the OMAT-FT checkpoint, Scratch, and Scratch-5% under the grouped protocol with three independent training initializations. On the controlled within-gap probe, using the current provisional DFT reference of 0.34 eV, the absolute fixed-path barrier errors are 0.47 ± 0.12 eV for FT-600K (OMAT-FT start), 1.20 ± 0.27 eV for Scratch, and

0.68 ± 0.20 eV for Scratch-5%. The single-checkpoint MACE-MPA-0 comparator gives 0.24 eV and is not included in any seed uncertainty. Thus, the historical 0.16 eV value does not survive the leakage-controlled, grouped multi-seed re-evaluation; we retain it only as a labeled frame-level historical point estimate and do not present it as the current FT-600K result.

The current DFT reference is the 0.34 eV candidate used consistently by this evaluator; its independent QE convergence qualification remains a separate provisional gate. The other evaluated paths are preserved in the machine-readable audit but are not pooled into the in-gap conclusion because their pathway class or reference identity is not jointly closed. FT-Multi$_T$ was not retrained under the grouped protocol in this round and is not assigned a grouped A2 result.

To make the distinction between static pointwise accuracy and process-conditioned configuration-space coverage explicit, we added two independently rendered configuration-space panels (Fig. 5 and Fig. 7), a separate four-model RMSE history (Fig. 6), and a readable stacked overview. The reference panel uses a PCA fit on grouped training data only and shows the complete 1,786/311/267 train/validation/test split with calibrated marginal densities and smooth two-dimensional density surfaces. The stored MLIP-generated 600 K MD rows comprise Foundation 4,961 rows with 3 SOAP-valid frames and 4,958 short-contact skips, Scratch 4,001 rows with 1,310 valid and 2,691 skipped, FT-600K 4,601 valid rows, and FT-Multi$_T$ 4,001 valid rows. These MD distributions are descriptive MLIP-only configuration-space diagnostics; the common MD velocity seed does not provide independent velocity replicas or calibrated uncertainty. The four-model RMSE figure reconstructs 87 original default-head validation checkpoints from the Rockfish JSONL logs, retaining the native training-history evidence while keeping it separate from grouped held-out test statistics. The binary stability panel and free-form diagnosis text were removed, and the stage-resolved failure/CAR diagnostic is provided separately in Fig. 8.

Table 2: Grouped-split and overlap audit summary. Barrier errors are absolute fixed-path errors on the controlled within-gap probe relative to the provisional 0.34 eV DFT candidate. Three-initialization entries report mean ± sample standard deviation; MPA0 is a single-checkpoint comparator. The overlap rows report the minimum training-set distance across the 25 audited images under the declared metrics.

Family/split	Test E RMSE (meV/atom)	Test F RMSE (meV/Å)	Within-gap |error| (eV)
OMAT-FT, grouped	40.7 ± 5.0	259.7 ± 5.8	0.47 ± 0.12
Scratch, grouped	30.5 ± 3.6	343.6 ± 35.9	1.20 ± 0.27
Scratch-5%, grouped	16.7 ± 9.2	341.5 ± 15.0	0.68 ± 0.20
MPA0, grouped comparator	38.7	278.5	0.24
Full grouped overlap	1,786 train frames	SOAP 5.4×10^{-4}	RMSD 2.06 Å
Scratch-5% overlap	89 train frames	SOAP 6.4×10^{-4}	RMSD 2.06 Å

Manuscript change.

Unchanged context: the overlap conclusion is bounded to the declared fine-tuning files and metrics, and the DFT candidate remains provisional.

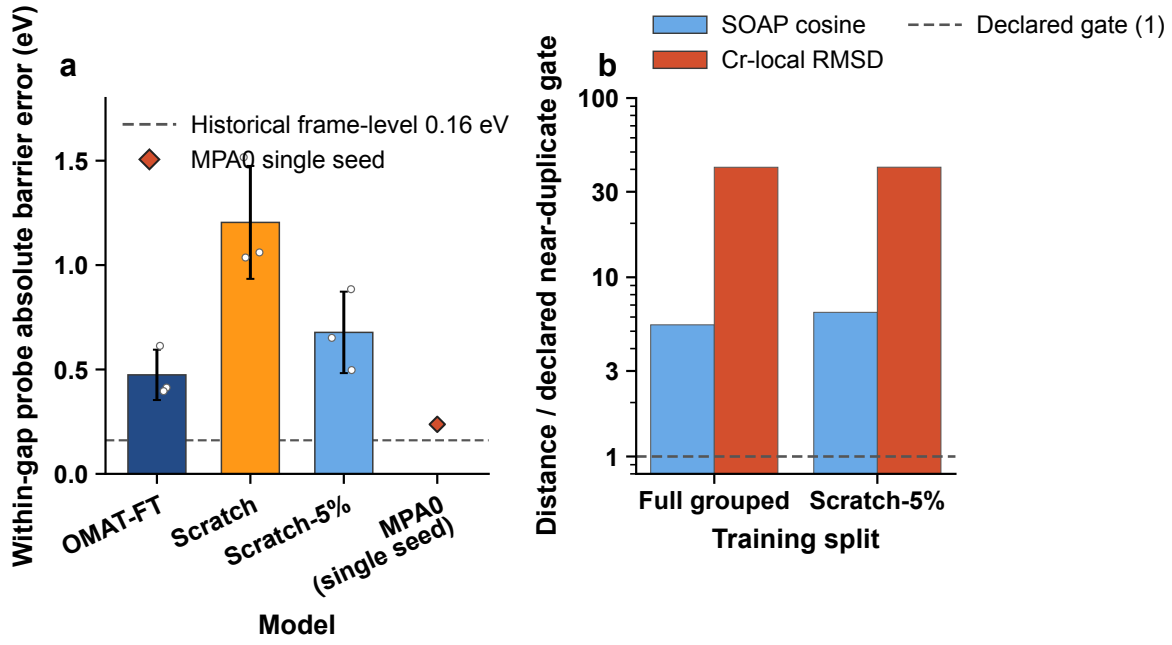


Figure 4: Leakage-controlled grouped retraining and overlap audit. (a) Absolute fixed-path barrier errors for the controlled within-gap probe across three independent training initializations, with open points for the individual runs and bars/whiskers for the mean ± sample standard deviation. MPA0 is shown as a single-checkpoint diamond. The dashed line is the historical frame-level 0.16 eV point estimate, which is not reproduced by the grouped FT-600K evaluation. (b) Minimum training-set SOAP and comparable Cr-local RMSD distances normalized by the declared near-duplicate gates. The dashed line at one is the declared near-duplicate gate; values above one lie outside that gate. RMSD uses only species-composition-compatible local environments, whereas SOAP covers all audited target images.

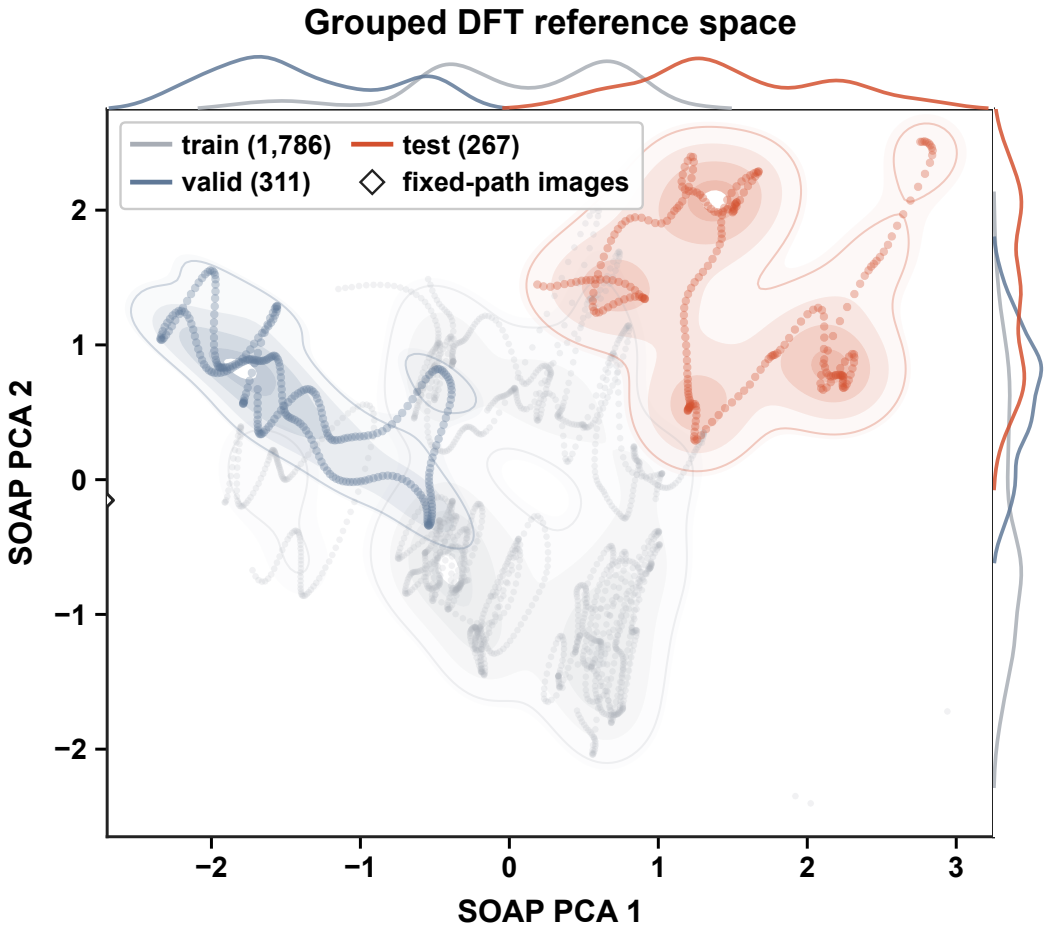


Figure 5: Grouped DFT reference space. SOAP/PCA is fitted on the trajectory-grouped training partition only and applied without refitting to 1,786 training, 311 validation, 267 test, and five fixed-path pathway-1–4 images. The training partition contains 1,783 periodic AIMD frames and three isolated-atom calibration frames; complete source AIMD trajectories are assigned to one partition. Soft two-dimensional density surfaces and calibrated marginal supports show the grouped split, while diamonds identify the fixed-path images. The latter are kept as a separate geometry class because the audited pathway-1–4 images contain 61 atoms, whereas the grouped AIMD cells contain 81–86 atoms. This panel is a descriptive representation diagnostic rather than a scalar OOD score.

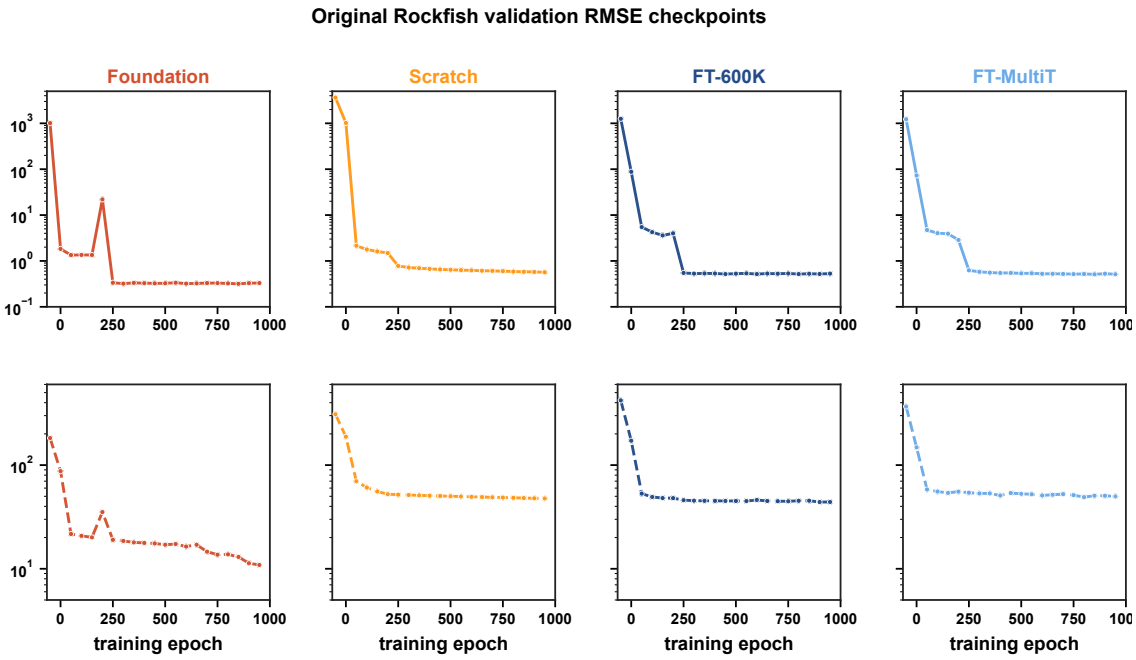


Figure 6: Original Rockfish validation RMSE checkpoints. The four columns follow the manuscript order Foundation, Scratch, FT-600K, and FT-MultiT. The upper row reports energy RMSE per atom and the lower row reports force RMSE, both converted from the original JSONL training logs to meV-based units. Each curve contains the unique default-head `mode=eval` checkpoints from the corresponding run 123 log; duplicate log checkpoints are retained in the raw table but omitted from the plotted trace. These are descriptive single-run validation histories from the original training campaigns, not grouped held-out test errors, seed uncertainty, or a universal model ranking. The source paths, SHA256 values, and model-lineage status are recorded in the accompanying RMSE manifest.

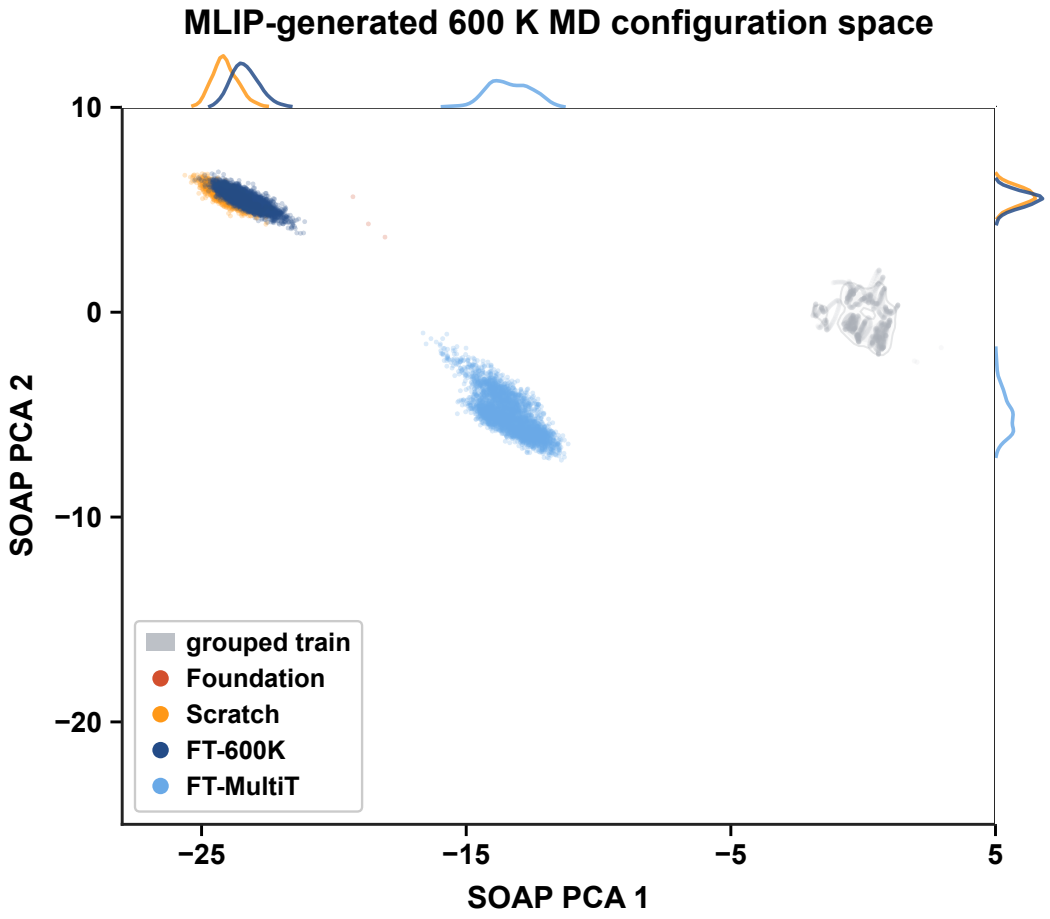


Figure 7: MLIP-generated 600 K MD configuration space. SOAP/PCA distributions from the stored MLIP-generated continuation segments are overlaid on the grouped-training density using the same PCA transform as panel (a). The stored rows are Foundation 4,961 (3 SOAP-valid and 4,958 short-contact skips), Scratch 4,001 (1,310 valid and 2,691 short-contact skips), FT-600K 4,601 valid, and FT-MultiT 4,001 valid. Short-contact rows remain in the raw inventory but are excluded from SOAP after the geometry-safety precheck. These clouds are MLIP-only distributional diagnostics rather than DFT errors; the common MD velocity seed means that the branches are not independent velocity replicas or calibrated uncertainty estimates.

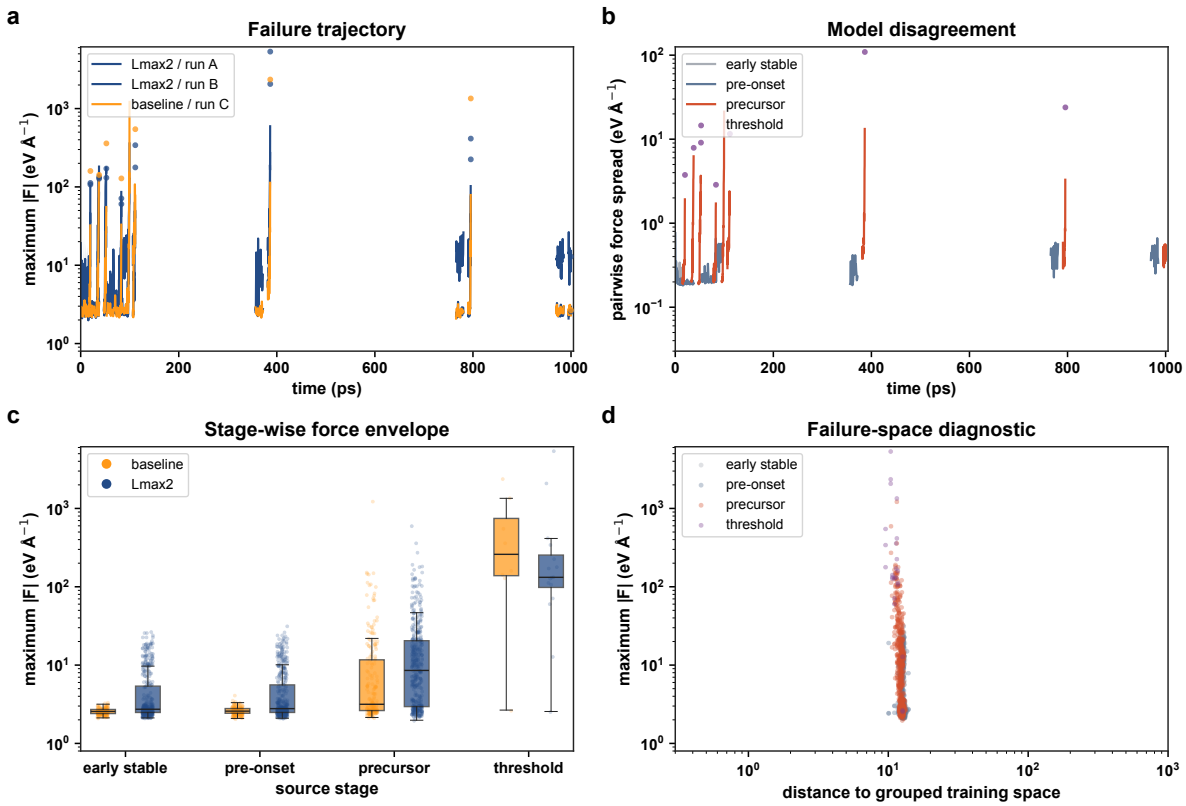


Figure 8: Stage-resolved MLIP-only failure/CAR diagnostic. (a) Maximum force along the 639-candidate failure trajectories for the baseline and $l_{\max} = 2$ branches. (b) Paired model force disagreement across the same stage windows. (c) Stage-wise force envelopes for the two model branches. (d) Maximum force as a function of distance to the grouped training representation. The stage-balanced samples are descriptive diagnostics of failure progression and model disagreement; they are not DFT errors, calibrated uncertainties, or a monotonic scalar OOD ranking.

Inserted/revised manuscript text.

This controlled within-gap probe provides a test of fixed-path energetic fidelity. Under the grouped split, train–test proximity audit, and three-initialization retraining, FT-600K (OMAT-FT start) has the lowest fixed-path barrier error (0.47 ± 0.12 eV), followed by Scratch-5% (0.68 ± 0.20 eV) and Scratch (1.20 ± 0.27 eV); the historical single-run frame-level estimate of 0.16 eV is not reproduced under this grouped, multi-run protocol.

Methods Sec. 2.1 and the SI retain the grouped split, SOAP/RMSD overlap metrics, complete grouped table, task-conditioned distribution diagnostics, and the separate MLIP-only failure/CAR panel.

A3. Uncertainty in kinetic results

Reviewer comment.

> *"Add uncertainty to the kinetic results: retrain at least Scratch and FT-600K with ≥3 seeds and report barrier-error mean +/- std (the 3-seed protocol already exists for Table S1). This simultaneously tests the 'chance' explanation for the scratch model's deep-penetration result—if the low error is seed-stable, 'chance' is refuted; if seed-dependent, it is supported. Either way, replace the current unfalsifiable assertion with evidence."*

Response. We agree. We completed grouped retraining for Scratch and FT-600K with three independent training initializations, and we now report variation across those runs rather than an assertion about chance. The controlled within-gap probe gives absolute fixed-path errors of 0.47 ± 0.12 eV for FT-600K, 1.20 ± 0.27 eV for Scratch, and 0.68 ± 0.20 eV for Scratch-5%. The historical frame-level 0.16 eV point estimate is not reproduced by this leakage-controlled grouped evaluation.

The same grouped files also allow the reviewer's chance explanation to be tested on the higher-distortion candidates without pooling them into a final hard-OOD ranking. For Scratch, one gap-to-QL/into-QL candidate gives an absolute error of 0.51 ± 0.11 eV across the three runs, a second gives 0.72 ± 0.55 eV, and a geometry-recheck candidate gives 1.77 ± 1.00 eV. The exact audit IDs and run-level values are retained in the SI table and raw CSV rather than used as the main narrative labels.

The corresponding FT-600K errors are 9.60 ± 0.57, 10.40 ± 0.44, and 9.70 ± 0.20 eV, respectively. Thus the lower error on the first gap-to-QL/into-QL candidate is not a single-run outlier within the three tested initializations, whereas the second candidate shows larger run-to-run spread and the geometry-recheck candidate remains both high-error and seed-sensitive. We replace the unfalsifiable phrase "by chance" with this class-aware conclusion: seed sensitivity is present, but it is not a uniform explanation for the higher-distortion results. This conclusion is conditional on the tested initializations and provisional DFT anchors; it does not establish physical accuracy on a hard-OOD pathway.

Evidence status. These grouped values are quantitative but conditional on provisional DFT candidates. The reported standard deviations are sample SDs across the three training seeds, not calibrated confidence intervals and not independent molecular-dynamics or velocity-seed replicas. Converged higher-distortion DFT references, final path classification, and a same-system self-relaxed calculation with DFT rescoring remain validation gaps.

Outstanding gate. A final higher-distortion/within-QL comparison requires the restarted QE reference and a path-resolved, seed-level error table. Until those are available, this reviewer request is addressed by reporting the completed within-gap uncertainty and explicitly withholding the higher-distortion model ranking.

Supporting protocol audit. The broader fixed-path and self-relaxed matrices are retained as descriptive supporting evidence with explicit model-lineage, force-pathology, and convergence flags. They are not pooled with the strict grouped uncertainty or used for a final higher-distortion ranking.

The complete fixed-path matrix contains 50 finite cells for the 61-atom system and 50 for the 81-atom system; 30 of the 100 cells carry an explicit force-pathology flag. The auxiliary 61-atom FT-600K and Scratch-Multi$_T$ subset contains 30 cells and 17 such flags. Across the audited within-gap and higher-distortion classes, the separate lineages show broad error ranges and retained force-pathology flags. These values document completed model/path diagnostics, but the 81-atom references retain high residual forces and the 61-atom higher-distortion references remain provisional.

The self-relaxed audit contains 18 optimizer-converged cells, 17 finite but non-converged cells, and 5 legacy diagnostics with incomplete convergence fields. These values are MLIP-native barrier proxies and the relaxed images have not been DFT-rescored, so they document protocol stability and failure modes rather than physical barrier accuracy.

Figure/table insert and evidence boundary. The response-letter insert is Fig. 9, together with a class-labeled summary table; audit IDs and run-level values remain in the SI and raw CSV. We also reuse the historical FIG-037 path-type summary, FIG-041–FIG-045 path-resolved profiles, FIG-060 NEB residual audit, FIG-194/FIG-195 fixed-path foundation profiles, and FIG-211 topology audit below. These pages expose the pathway classes and reference-quality limitations that motivate the class-aware interpretation; they do not replace the new seed-resolved figure or create a pooled hard-OOD ranking. The SI additionally includes the fixed-path matrix figure (50/50 finite 61-atom cells and 50/50 finite 81-atom cells, with retained force-pathology marks) and the self-relaxed matrix figure (18 converged, 17 finite-but-non-converged, and 5 legacy diagnostic cells). The latter is explicitly captioned as an MLIP-native path-generation diagnostic; neither matrix figure is used as a DFT barrier ranking or as a substitute for same-system DFT rescoring.

Reused path-level context for A3. These historical pages make the pathway-class boundary and the path-to-path variability visible. They are retained as descriptive/audit evidence; the quantitative uncertainty claim remains the grouped three-initialization result in Fig. 9.

Table 3: Class-resolved kinetic errors. Entries are absolute fixed-path barrier errors relative to the current pathway-specific DFT candidate. The table summarizes three independent training initializations; run-level observations and audit IDs remain in the accompanying CSV and SI audit table. Higher-distortion references are provisional and one candidate requires a geometry recheck.

Pathway class	Model	Mean ± SD (eV)
Gap-to-QL/into-QL candidate A	Scratch	0.51 ± 0.11
Gap-to-QL/into-QL candidate A	FT-600K	9.60 ± 0.57
Gap-to-QL/into-QL candidate B	Scratch	0.72 ± 0.55
Gap-to-QL/into-QL candidate B	FT-600K	10.40 ± 0.44
Controlled within-gap probe	Scratch	1.20 ± 0.27
Controlled within-gap probe	FT-600K	0.47 ± 0.12
Controlled within-gap probe	Scratch-5%	0.68 ± 0.20
Geometry-recheck candidate	Scratch	1.77 ± 1.00
Geometry-recheck candidate	FT-600K	9.70 ± 0.20
Additional within-gap candidate	Scratch	0.33 ± 0.06
Additional within-gap candidate	FT-600K	0.21 ± 0.18

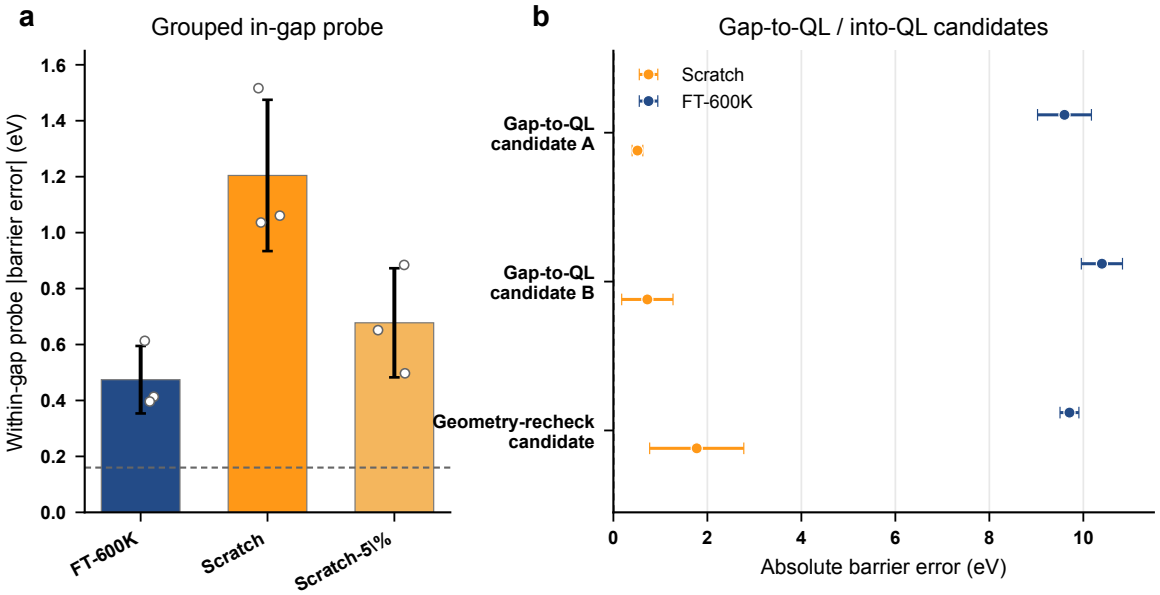


Figure 9: Seed-resolved kinetic uncertainty. (a) Grouped three-run absolute fixed-path errors on the controlled within-gap probe; open points are the individual training initializations and the dashed line marks the historical 0.16 eV point estimate, which is not reproduced. (b) Means and sample standard deviations for representative gap-to-QL/into-QL candidates, including a geometry-recheck case. The first candidate is comparatively stable across the tested initializations, whereas the other candidates are more seed-sensitive. All higher-distortion DFT references remain provisional; exact audit IDs are given in the SI table.

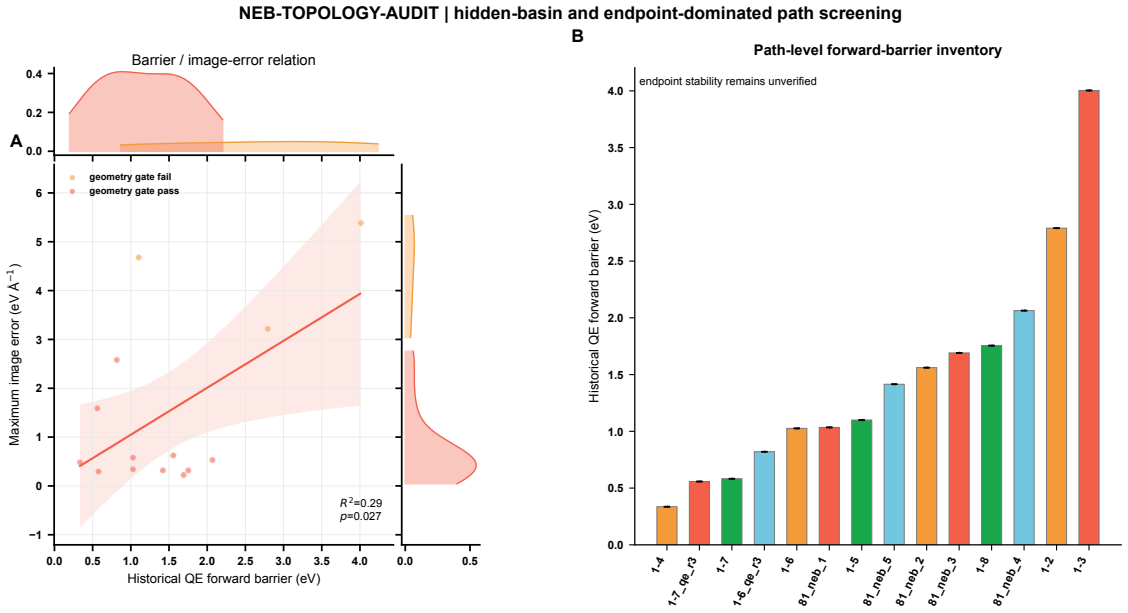


Figure 10: Reused path-topology audit for the A3 boundary. This existing review-book asset (FIG-211) shows the relation between image-error diagnostics and the historical/candidate path inventory, including endpoint-dominated and hidden-basin candidates. It is included as a provenance and pathway-classification audit, not as a replacement for the grouped seed-resolved uncertainty figure. The 0.336 eV path-1–4 value remains a candidate fixed-path reference, and the figure does not certify endpoint stability, NEB convergence, or a final higher-distortion ranking.

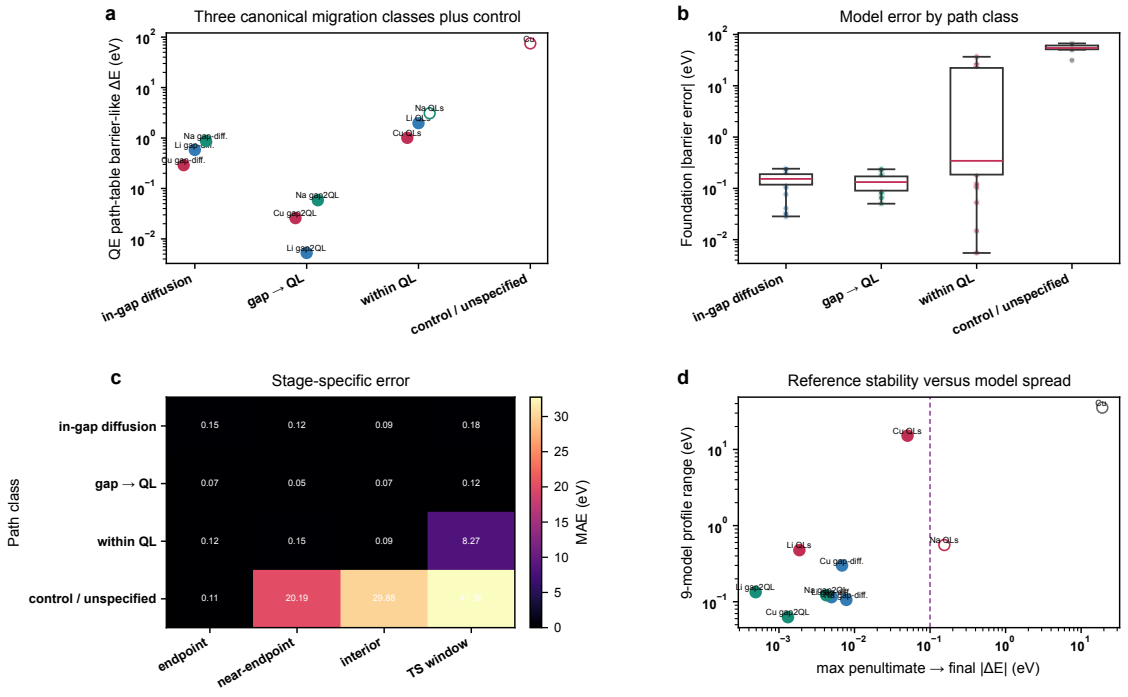
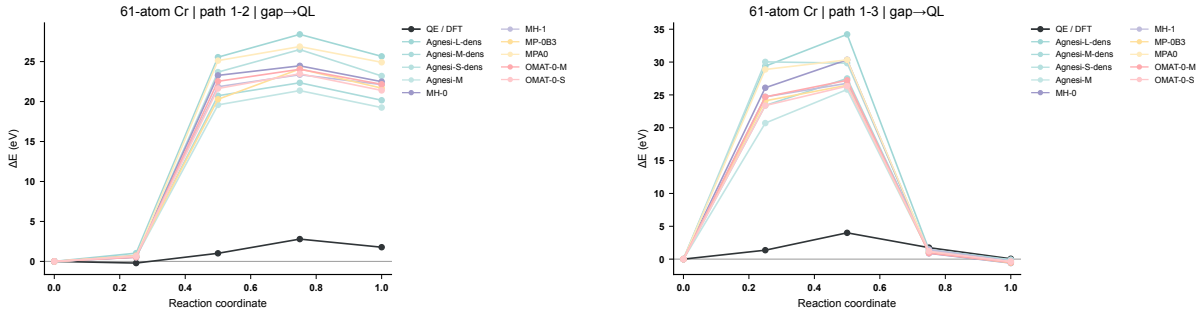


Figure 11: Reused FIG-037 path-type summary. The historical summary separates the controlled within-gap/interpolative probe from gap-to-QL/into-QL and other higher-distortion candidates. It is included to document the classification boundary used in the revised A3 analysis. The categories and DFT references remain audit-level/provisional where flagged and are not pooled into a final hard-OOD ranking.



(a) Path 1–2. **(b)** Path 1–3.

Figure 12: Reused higher-distortion path profiles (FIG-041/FIG-042). These fixed-image profiles provide path-resolved context for the two deep-penetration candidates. They are historical model/path diagnostics rather than the new grouped seed statistics; the DFT anchors and pathway labels remain provisional, so these panels are descriptive and are not used to claim a final ranking.

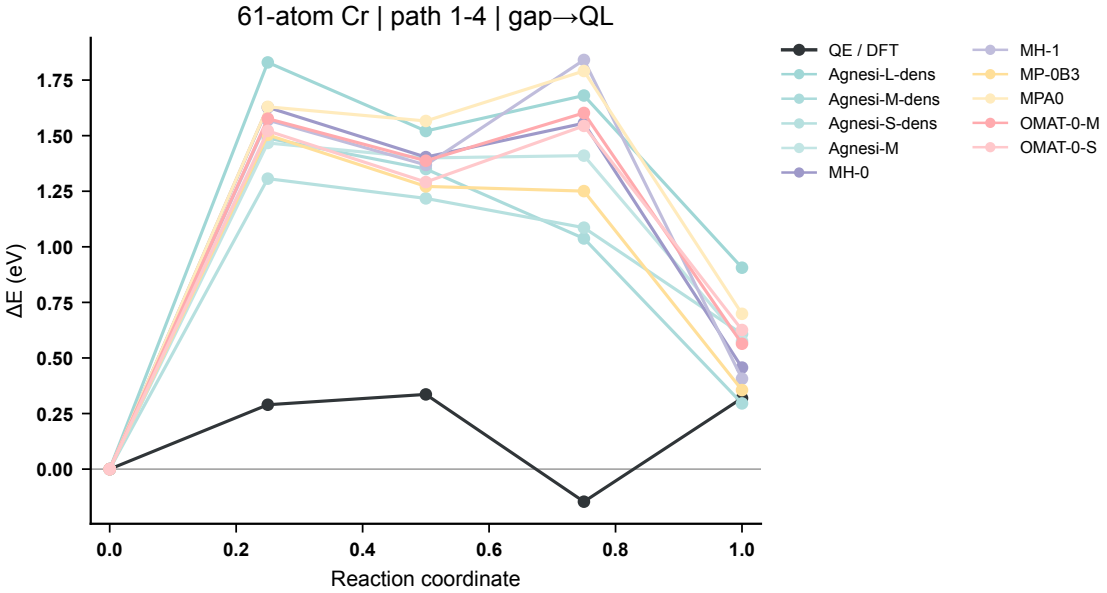
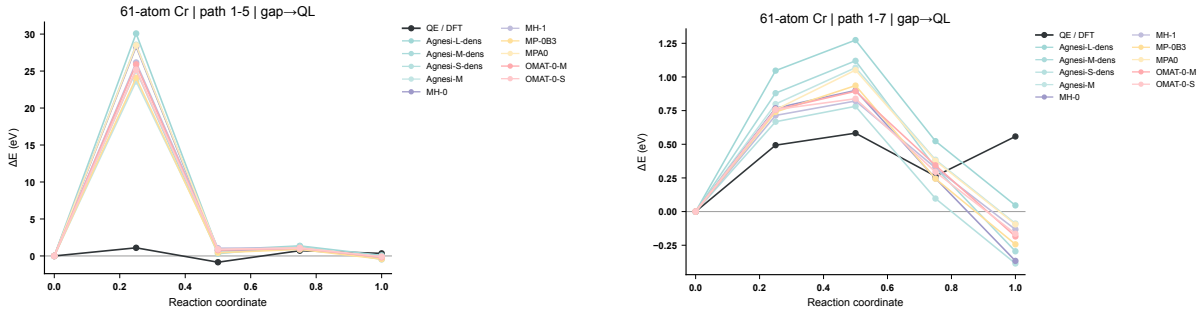


Figure 13: Reused controlled path-1–4 profile (FIG-043). The historical export labels this panel "gap→QL", whereas the current grouped audit treats the path-1–4 record as the controlled within-gap reference; this label reconciliation is retained in the provenance notes. The profile visualizes the common DFT-defined images and model-specific energy assignment, while the current manuscript reports the newer seed-resolved absolute errors. The DFT barrier remains a provisional candidate and the profile is not a self-relaxed NEB result.



(a) Path 1–5; geometry recheck. **(b)** Path 1–7; in-gap candidate.

Figure 14: Reused geometry/classification boundary profiles (FIG-044/FIG-045). The path-1–5 panel is retained with its geometry-recheck status, while path 1–7 is retained as a separate in-gap candidate. The panels show why the current analysis keeps path classes separate and labels the additional rows descriptive rather than pooling them into a single uncertainty estimate.

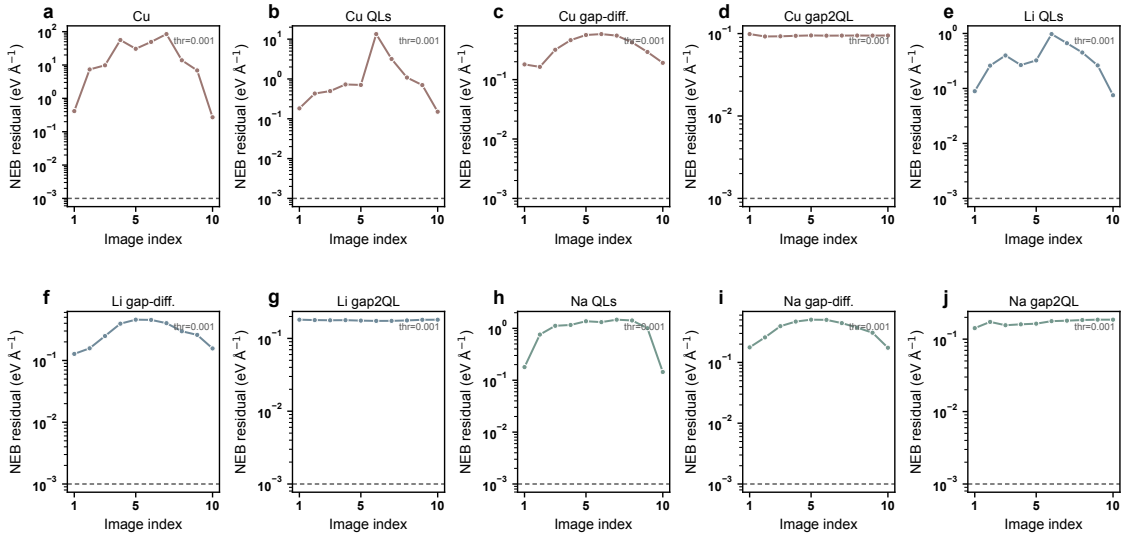
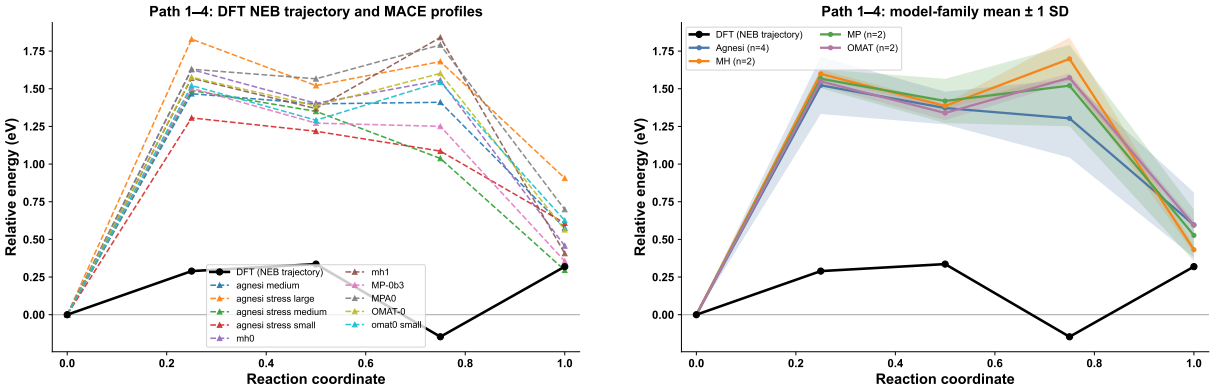


Figure 15: Reused FIG-060 NEB residual audit. This historical audit makes the residual-force and convergence limitations visible for the DFT path records. It supports the decision to keep the DFT anchors provisional and to withhold a final higher-distortion ranking; it is not itself a converged DFT barrier result.



(a) FIG-194: ten-checkpoint profiles. **(b)** FIG-195: family aggregation.

Figure 16: Reused foundation fixed-path profiles for A3. These existing pages show the matched single-checkpoint foundation profiles and family-level spread on path 1–4. They support the distinction between pointwise/static agreement and fixed-path kinetic calibration, but the family bands are sub-model variation rather than seed uncertainty and the reference path remains provisional.

Manuscript change.

Unchanged context: the revision remains a one-system case study, and the DFT/pathway gates are still explicitly open for the higher-distortion family. The blue callouts reproduce the wording that is now present in the main Results and SI sections.

Inserted/revised manuscript text.

Under the grouped-split, three-initialization re-evaluation, the absolute fixed-path barrier error on the controlled probe relative to the provisional DFT estimate (0.34 eV) is 0.47 ± 0.12 eV for FT-600K fine-tuned from the OMAT-FT checkpoint, 1.20 ± 0.27 eV for Scratch, and 0.68 ± 0.20 eV for Scratch-5%; a single-checkpoint MACE-MPA-0 comparator gives 0.24 eV. FT-600K has the lowest reported error on this controlled probe. FT–MultiT was not retrained under the grouped protocol in this round.

Inserted/revised manuscript text.

These values replace a qualitative "by chance" explanation with evidence, but they do not establish a final hard-OOD ranking because the candidate DFT references and pathway classes are not uniformly closed.

The SI section "Seed-aware kinetic uncertainty and protocol audit" adds the complete audit-ID table, the grouped seed figure, the fixed-path/self-relaxed protocol matrices, and the reused path-topology/profile assets. Their captions retain the labels descriptive, provisional, or validation gap wherever the corresponding scientific gate is not closed.

A4. Scope of generalizability claims

Reviewer comment.

> *"Rescope the generalizability claims: change 'establishes ... a generalizable standard' to a candidate framework demonstrated on one system, and move 'broader principles' to a hedged outlook—or add one additional material/dopant (even a cheap second system, e.g., a different transition-metal dopant in the same host) as minimal cross-validation."*

Response. We agree and have adopted the requested narrower claim. The manuscript now describes MigrationBench as a candidate task-conditioned model-selection framework demonstrated on one Cr-doped Sb_2Te_3 system and the MACE model family. It no longer presents the workflow as a generalizable standard or a universal ordering of specialist and foundation MLIPs. Broader principles are stated as hypotheses that require validation across additional dopants, hosts, defect environments, and MLIP architectures.

We also audited an existing cross-element diagnostic to determine whether it could support a minimal second-system claim. Three Cr-trained broad-roster checkpoints, all from the recorded final epoch, match exactly across the 90 state-dict tensors and give validation energy RMSEs of 11.33, 12.16, and 12.59 meV/atom and force RMSEs of 203.62, 208.76, and 222.02 meV/Å. The models were evaluated on ten archived Cu/Li/Na paths in the same Sb_2Te_3 host. All images were finite and passed image-count, endpoint-identity, and PBC checks, and 8/10 passed the package's < 0.10 eV final-table screen. However, all ten DFT records retain SCF-convergence

warnings and therefore fail the scientific DFT gate. We do not convert these screens into DFT barrier errors, transfer gains, or a second-material validation claim.

On the six gap-like paths, the checkpoint-to-checkpoint maximum relative-energy diagnostics are 0.06 ± 0.01 eV for Cu gap-diffuse, 0.02 ± 0.01 eV for Cu gap2QL, 0.10 ± 0.04 eV for Li gap-diffuse, 0.07 ± 0.03 eV for Li gap2QL, 0.30 ± 0.17 eV for Na gap-diffuse, and 0.12 ± 0.04 eV for Na gap2QL. These are MLIP diagnostic snapshot variations, not barrier-error uncertainties, because the corresponding DFT references are not scientifically converged. The result supports reproducibility and evidence-gating, while leaving cross-element transfer as an explicit validation target.

Table 4: A4 cross-element diagnostic. The Cr-trained broad-roster models were evaluated on archived Cu/Li/Na paths in the same host. All ten DFT records retain SCF-convergence warnings and therefore fail the scientific DFT gate. The final-column diagnostic is the mean ± sample SD of the maximum MLIP relative energy across three checkpoints; it is not a DFT barrier error.

Element	Path	DFT gate	< 0.1 eV screen	Max relative energy (eV)	Seed range (eV)
Cu	`Cu_neb`	failed	no	123.43 ± 3.95	8.45
Cu	`Cu_neb-QLs`	failed	yes	1.15 ± 0.03	0.08
Cu	`Cu_neb-gap-diffuse`	failed	yes	0.06 ± 0.01	0.02
Cu	`Cu_neb-gap2QL`	failed	yes	0.02 ± 0.01	0.03
Li	`Li_neb-QLs`	failed	yes	1.14 ± 0.02	0.05
Li	`Li_neb-gap-diffuse`	failed	yes	0.10 ± 0.04	0.09
Li	`Li_neb-gap2Ql`	failed	yes	0.07 ± 0.03	0.06
Na	`Na_neb-QLs`	failed	no	1.62 ± 0.10	0.25
Na	`Na_neb-gap-diffuse`	failed	yes	0.30 ± 0.17	0.39
Na	`Na_neb-gap2QL`	failed	yes	0.12 ± 0.04	0.08

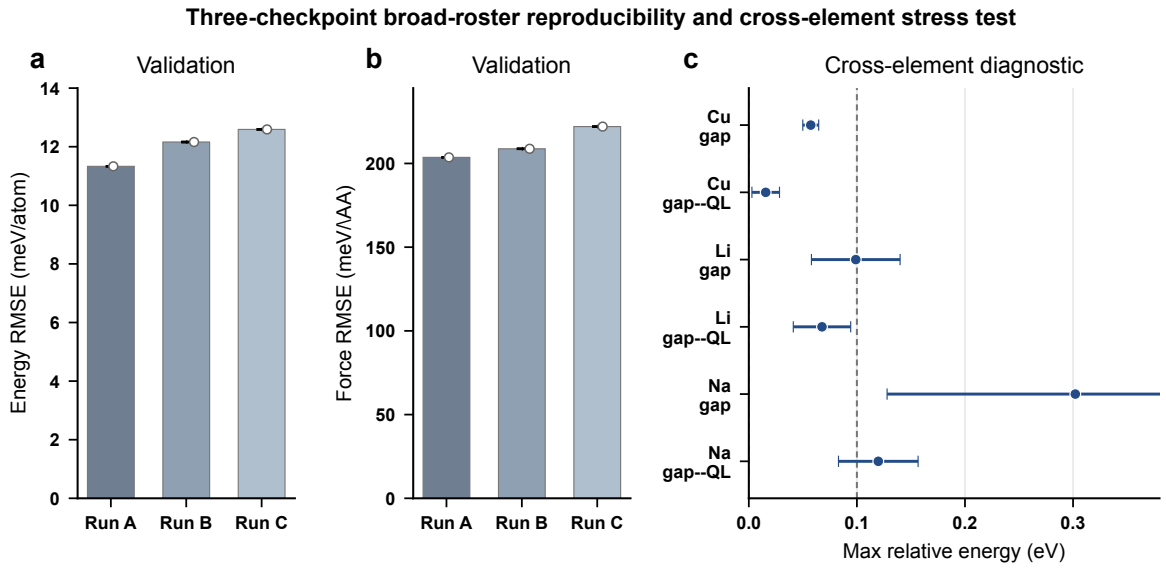


Figure 17: Cross-element diagnostic used to rescope A4. (a,b) Validation RMSEs for three broad-roster checkpoints. (c) Checkpoint-to-checkpoint maximum relative-energy diagnostics on six gap-like Cu/Li/Na paths; the dashed line is the package's 0.10 eV triage screen. All ten archived DFT path records retain SCF-convergence warnings. The panels therefore document reproducible diagnostic behavior, not cross-element DFT barrier accuracy.

Reused framework and cross-element context for A4. These existing review-book assets are included because they already visualize the task-conditioned evaluation ladder and the recorded cross-element seed/validation diagnostics. They are supporting context, not evidence of universal transfer.

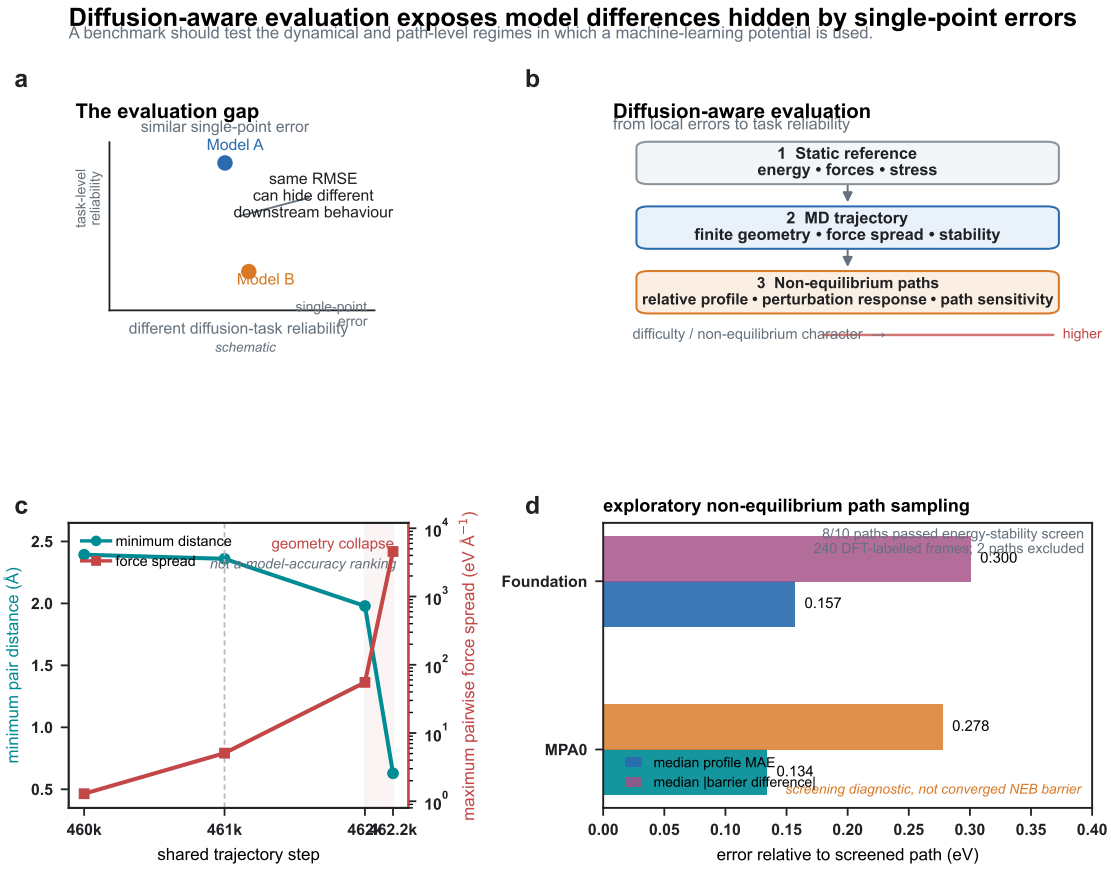


Figure 18: Reused FIG-001 diffusion-aware evaluation overview. This earlier overview connects model selection to increasingly demanding dynamic and pathway probes. It is reused here as visual support for the candidate task-conditioned framework demonstrated on one Cr-doped Sb_2Te_3 system; it does not establish a universal ranking or cross-material generalization.

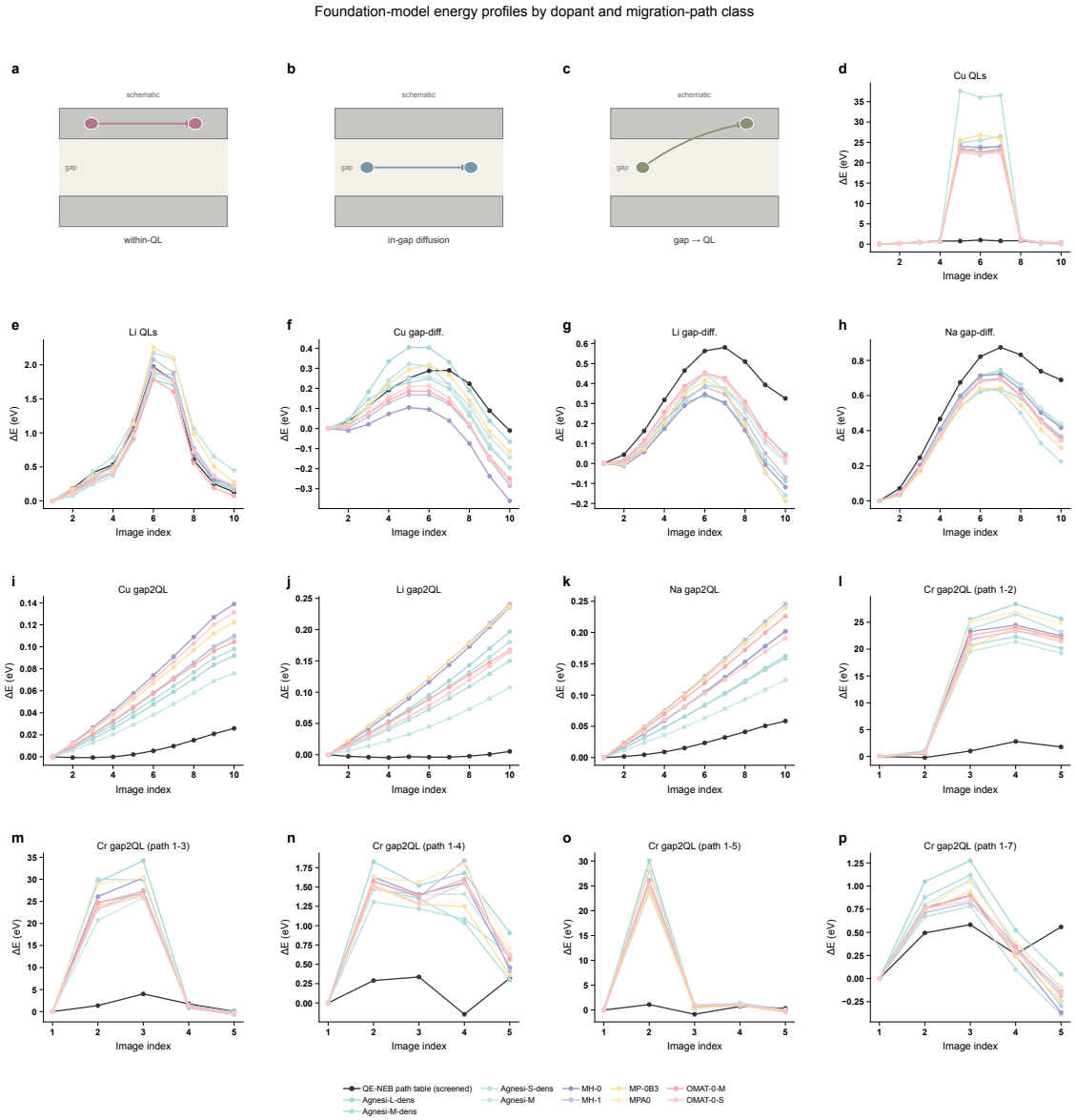
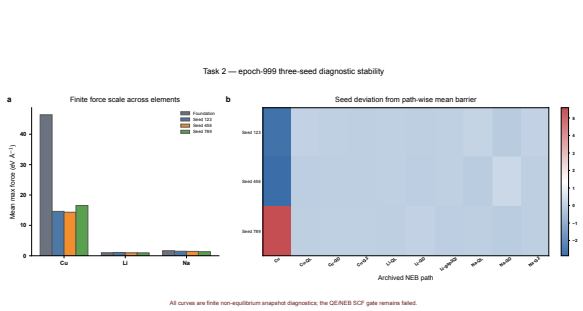
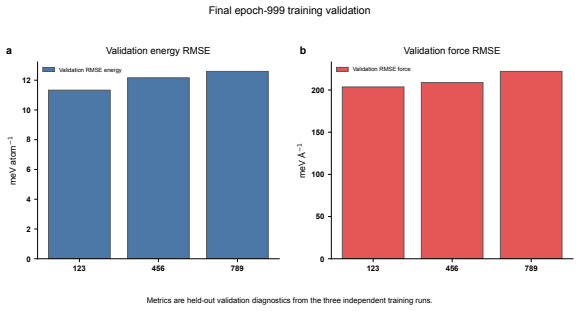


Figure 19: Reused FIG-057 task-type matrix. The historical matrix separates static, equilibrium, fixed-path, self-relaxed, and broader pathway diagnostics. It is used to reinforce the current evaluation ladder and the distinction between a common DFT path and model-in-the-loop path generation. Historical labels are retained as audit context and are not interpreted as a universal difficulty scale or model ranking.



(a) FIG-031: three-seed stability.



(b) FIG-032: three-seed validation.

Figure 20: Reused cross-element diagnostic pages (FIG-031/FIG-032). These existing pages provide descriptive seed-resolved validation and roster coverage for the Cu/Li/Na stress test. All ten archived DFT path records retain SCF-convergence warnings, so the panels support reproducibility and evidence-gating discussion but do not establish cross-element barrier accuracy or a second-material validation claim.

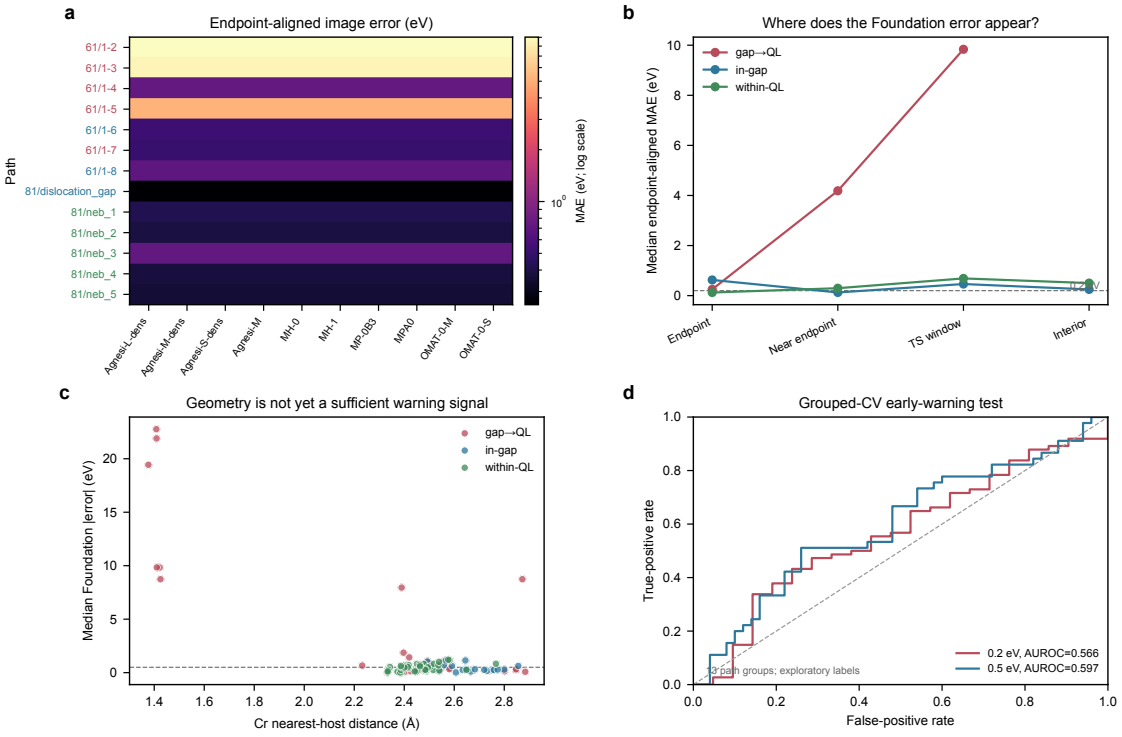


Figure 21: Reused FIG-052 unified diagnostic framework. This exploratory Cr-61/Cr-81 page is included as a visual provenance aid for separating system identity, path class, and evaluation operator. Because it mixes atom counts and path families, it is not a pooled quantitative result and is not used to support a universal generalizability claim.

Manuscript change.

Unchanged context: the benchmark remains a candidate framework demonstrated on one Cr-doped Sb_2Te_3 system; the Cu/Li/Na audit is retained only as a descriptive scope diagnostic.

Inserted/revised manuscript text.

This work demonstrates the candidate framework for Cr-doped Sb_2Te_3 within the current MACE model family (Figure 1). The material supplies a compact task ladder: near-equilibrium configurations, a controlled within-gap migration path, gap-to-QL/into-QL and within-QL candidates, and collective interlayer sliding. We evaluate four training strategies—training from scratch, zero-shot foundation inference, single-temperature fine-tuning, and multi-temperature fine-tuning—using near-equilibrium errors, fixed-path migration energies, self-relaxed NEB behavior, MD stability under the tested protocol, and exploratory representation diagnostics. The benchmark separates two questions that are often conflated: whether a model is calibrated on a common reference path, and whether its own forces can stably generate a path or trajectory. It is therefore a case-study framework rather than a universal ranking of MLIP architectures or materials.

Inserted/revised manuscript text.

The cross-element broad-roster audit provides a reproducibility and evidence-gating diagnostic, but its ten Cu/Li/Na DFT path records retain SCF-convergence warnings and therefore do not constitute transfer validation. Accordingly, the generality of the task-conditioned framework beyond this Cr-doped Sb_2Te_3 case remains an open validation question.

The SI section "Cross-element diagnostic and scope of generalizability" adds the cross-element table and figure, the reused FIG-001/FIG-031/FIG-032/FIG-052/FIG-057 context pages, and the explicit distinction between descriptive diagnostics, provisional references, and completed transfer validation.

A5. Scratch-5% control

Reviewer comment.

> *"Add a Scratch-5% control (same ∼1,000 configurations as FT-600K, random initialization) to disentangle foundation-pretraining benefits from data-volume effects."*

Response. We agree that a random-initialization low-data control is the correct way to separate a data-volume effect from a foundation-model initialization effect. We implemented this control within the leakage-controlled trajectory-grouped protocol and make one protocol distinction explicit. The historical FT-600K description referred to a frame-level subset of approximately 1,000 configurations, whereas the revised FT-600K comparison uses the full grouped training partition (1,786 configurations). Selecting exactly 1,000 frames while preserving complete source trajectories would fragment the grouped split and would reintroduce the train/test-overlap concern. We therefore do not relabel the historical frame-level subset as the present control: the reproducible current Scratch-5% control contains 89 grouped training configurations, with the same 311 validation and 267 test configurations, the same declared MACE architecture and training settings, random initialization, and three independent training initializations. It is a leakage-controlled low-data control, but not an exactly matched 1,000-configuration causal ablation.

Grouped test RMSE: 16.7 ± 9.2 meV/atom (energy), 341.5 ± 15.0 meV/Å (force). Controlled within-gap absolute fixed-path barrier error: 0.68 ± 0.20 eV, versus 0.47 ± 0.12 eV for FT-600K (OMAT-FT start) and 1.20 ± 0.27 eV for full Scratch.

The comparison is informative in two ways. First, the low-data Scratch-5% model does not simply degrade on every pointwise metric: its energy RMSE is lower than full Scratch (16.7 versus 30.5 meV/atom), while its force RMSE is essentially the same (341.5 versus 343.6 meV/Å). Second, on the controlled within-gap probe, its barrier error lies between FT-600K and full Scratch (0.68 versus 0.47 and 1.20 eV, respectively). The same low-data model therefore looks competitive on static pointwise metrics but is not the best process-level model, which is precisely the task-conditioned distinction the benchmark is intended to expose. The ordering is consistent with useful target adaptation from the foundation initialization under moderate distribution shift, but it does not isolate a pure pretraining benefit because the present grouped control has 89 rather than exactly 1,000 training configurations and the FT-600K comparison uses the full grouped training partition. We therefore present this as evidence supporting the task-conditioned interpretation, not as a causal estimate of the pretraining contribution.

For completeness, we also processed the existing path-level raw data rather than discarding it. Across the additional within-gap and gap-to-QL/into-QL candidates, the Scratch-5% absolute errors include 0.33 ± 0.31 eV and 1.22 ± 0.66 eV, while the corresponding FT-600K values include 9.60 ± 0.57 eV and 10.40 ± 0.44 eV. The full path-resolved table, including the controlled within-gap row (0.68 ± 0.20 eV for Scratch-5%), is added to the SI with internal audit IDs, DFT candidate values, and pathway classes. These additional rows show that the low-data control is pathway-dependent, but the candidate DFT anchors and class assignments are not uniformly closed, so they are not pooled into a final higher-distortion ranking.

The existing grouped-retraining audit figure is reused as the visual insert: its individual-run points already show the Scratch-5% control beside FT-600K and Scratch on the controlled within-gap probe, while the new SI table exposes the additional path-resolved raw evidence. The exact same-∼1,000 grouped control, a grouped FT-MultiT comparison, and converged seed-level higher-distortion references remain validation gaps.

Manuscript change.

Unchanged context: the low-data control is informative but is not presented as a causal estimate of pretraining or as an exactly matched 1,000-configuration ablation.

Inserted/revised manuscript text.

The 600 K fine-tuned model is closer to the controlled within-gap migration probe than the broader multi-temperature model in this case study. The grouped-split SOAP/RMSD audit finds no fixed-path-image overlap with either the full training set or the Scratch-5% subset. The Scratch-5% control gives a within-gap error of 0.68 ± 0.20 eV, between FT-600K (OMAT-FT start; 0.47 ± 0.12 eV) and Scratch (1.20 ± 0.27 eV). This ordering is consistent with a useful foundation initialization under moderate distribution shift, but the 89-frame grouped control does not isolate a pure pretraining effect at exactly matched approximately 1000-frame data volume. FT–MultiT was not retrained under grouped protocol in this round, and the higher-distortion outputs are retained as class-resolved descriptive diagnostics in the Supplementary Information rather than pooled into a final ranking.

Sec. 2.1, Sec. 3.3.3, the grouped RMSE/barrier table, and the class-labeled SI path table retain

the exact data-volume and provisional-reference boundary.

A6. Interpretability claims: t-SNE/PHATE and SHAP

Reviewer comment.

> *"Soften the interpretability claims: reframe t-SNE/PHATE conclusions as 'consistent with' rather than 'direct mechanistic explanation,' report projection sensitivity to perplexity/seed, and compute at least one separation metric in the original descriptor space. For SHAP, report per-model surrogate R^2, rescope 'the model internalizes X' to 'the surrogate's error predictions are most sensitive to X,' and (ideally) add a perturbation test on MACE's own inputs for the dominant Cr-Cr channel."*

Response. We agree with the reviewer and have narrowed the claims throughout the manuscript. t-SNE and PHATE are now described as descriptive projections that are consistent with representational differences, not as direct mechanistic explanations. The historical feature matrix contained 6,192 columns, including one asserted all-zero force-sensitivity placeholder; the revision-facing analysis excludes that inactive column and uses 6,191 descriptors across 404 trajectory frames. The legacy projection scripts were audited and quarantined from the canonical reproduction release, and the retained analysis reports both projection-space sensitivity and separation metrics in the original descriptor space, together with the scaling caveat needed to interpret those metrics.

The local re-analysis gives the following values:

Table 5: Audited separation metrics across projected and descriptor spaces. The t-SNE sensitivity range is driven by perplexity in this implementation: at fixed perplexity, the nominal seeds 0–2 give bitwise-identical silhouette values under the locked PCA initialization. The projected scores are therefore descriptive diagnostics rather than independent estimates of model uncertainty.

Space	Overall silhouette	Notes
Old 2D t-SNE, $p = 30$, seed 42	0.41	Original projection-space baseline
2D t-SNE, $p \in \{5, 30, 50\}$, 3 nominal seeds	0.40 ± 0.03	Range 0.36–0.42; trustworthiness 0.963–0.973
PHATE 2D, 3 nominal seeds	0.47 ± 0.00	Range 0.47265–0.47277
Original 6191-d, Euclidean	0.33	Primary descriptor-space metric
Original 6191-d, cosine	0.28	Alternative descriptor-space metric
Original 6191-d, z-scored Euclidean	-0.02	Demonstrates scaling sensitivity
PCA 95%, z-scored Euclidean	-0.03	Conservative high-dimensional check

For SHAP, we now distinguish in-sample fit from cross-validated surrogate fidelity. The corrected workflow joins 150 frames by explicit `image_index` and uses 390 SOAP descriptors plus 17 structural descriptors. The revised 5-fold cross-validation metrics are:

Table 6: Cross-validated surrogate-model performance used for SHAP analysis.

Model	5-fold CV R^2	5-fold CV MAE
FT-600K	0.98	0.04
FT-MultiT	0.88	0.03
Scratch	0.97	0.03

These values support the use of the surrogate for the selected error target, but they do not

establish a causal attribution within MACE. The revised wording is therefore "the surrogate's error predictions are most sensitive to ...", rather than "the model internalizes ...". The corrected descriptor-to-physical-feature audit also changes the interpretation of the former Cr–Cr label: the largest FT-MultiT contribution is `Cr-Sb_n43_l0`; Cr–Cr descriptors occur in the FT-600K and Scratch rankings but are not a stable dominant channel across the three models, and the earlier `Cr-Cr_n13_l3` label is not retained as a reliable physical label.

We also completed a complementary perturbation check on native MACE endpoint relaxations. The fixed-cell campaign contains 846 formal tasks (six Cr/V/Mn–Sb_2Te_3/Bi_2Te_3 systems, 47 sites, and three arms: Base, a +0.20 Å dopant displacement, and a norm-0.20 Å vector displacement). All 846 tasks passed the finite-value, topology, minimum-distance, displacement, and convergence gates. For Cr@Sb_2Te_3, the median maximum displacement is 0.02/0.20/0.19 Å for Base/Perturb-A/Perturb-B; for Cr@Bi_2Te_3, it is 0.47/0.49/0.47 Å. Final-force medians remain close to 0.03 eV/Å. This response is host- and dopant-dependent and is useful as a native-MACE local-stability diagnostic, but it is not an isolated Cr–Cr SOAP-channel perturbation, not a DFT or magnetic validation, and not a causal test of the SHAP ranking. We therefore retain the exact isolated Cr–Cr input perturbation as a validation gap and make no causal Cr–Cr claim.

Evidence status. The projection and original-space numbers are audited descriptive evidence: the raw Euclidean score is scale-sensitive because the descriptor vector mixes total energy, forces, RDF features, and SOAP components. The overall silhouette decreases from 0.33 in the raw space to −0.02 after z-scoring and to −0.03 for the 95% z-scored PCA space, so the result supports distinguishability of the recorded model outputs but not disjoint local-geometry manifolds or a physical mechanism. The SHAP values are surrogate-fidelity evidence from a corrected 150-frame workflow and remain provenance-partial with respect to the historical end-to-end export; their interpretation is descriptive, not causal. The 846-task perturbation campaign is completed native-MACE endpoint evidence, while the isolated Cr–Cr perturbation requested by the reviewer remains a validation gap.

Figure/table insert and captions. We retain the existing Plot-Atlas-style latent-robustness figure and add the audited original-space table; their captions state that projection scores are descriptive and that the raw-space score is scale-sensitive. The revised Fig. 6 caption reports the corrected top feature and states that SHAP explains surrogate error predictions. The SI endpoint appendix retains the "Perturbation response across systems and endpoints" figure, with bootstrap intervals explicitly defined as summaries of the observed endpoint sample rather than uncertainty in the MACE potential.

Manuscript change.

Unchanged context: the representation and surrogate results remain descriptive, surrogate-scoped, and separate from causal claims about the MLIP.

Inserted/revised manuscript text.

Sec. 3.4 and the Fig. 5 caption now report the 0.36–0.42 t-SNE range, the PHATE sensitivity, the original-space metrics, and the scale caveat. Sec. 3.5 and the Fig. 6 caption report the per-model 5-fold surrogate R^2 values, the corrected descriptor labels, and the non-causal scope. The SI adds the audited representation/surrogate subsection, the original-space table, and the completed endpoint perturbation appendix. The direct isolated Cr–Cr perturbation is marked as a validation gap rather than being implied by the complementary endpoint campaign.

A7. Related work and distinction from our earlier report

Reviewer comment.

> *"Position against the nearest neighbors: cite and discuss arXiv:2510.05020 (MACE fine-tuning for Li-ion diffusion), the foundational-MLIP migration-barrier evaluation (RSC Digital Discovery d5dd00534e), and arXiv:2405.07105 (systematic softening—currently invoked as if novel). Add one paragraph explicitly stating this manuscript's delta over the authors' own arXiv:2509.00090 [46]."*

Response. We agree. We have added a dedicated related-work comparison in the Introduction and clarified the interpretation of the softening result in Sec. 3.2. Alghamdi et al. [1] compare MACE-MPA-0 and fine-tuned MACE for Li diffusion in LiF using nanosecond-scale molecular dynamics and Arrhenius observables; their reported activation energies are approximately 0.22 eV for MACE-MPA-0, 0.20 eV after fine-tuning on 300 configurations, and 0.24 eV for the DeePMD reference. Bheemaguli et al. [2] evaluate zero-shot foundation MLIPs coupled to NEB over 574 battery migration paths and separate barrier accuracy, ionic-conductor classification, and local transition-path geometry. Deng et al. [3] establish systematic underprediction of energy, force, and potential-energy-surface curvature in universal MLIPs on out-of-distribution tasks and show that sparse high-energy fine-tuning can correct part of this bias. We now cite systematic softening as established prior work, not as a phenomenon first introduced here; our contribution is to test its task-conditioned consequences in a controlled doped layered-material benchmark.

We also state explicitly that arXiv:2509.00090 was disseminated as a non-archival AI4Mat workshop report [4]. The present manuscript is an expanded journal study, not a duplicate publication. Relative to that report, it adds a formal measurement-operator distinction between fixed-path single-point scoring and self-relaxed model-in-the-loop NEB, separate in-gap and higher-distortion pathway classes, a complementary nonlocal interlayer-sliding probe, matched fixed-path evaluation of ten foundation checkpoints, representation and surrogate-error diagnostics, provenance-aware grouped splitting and nearest-neighbor overlap audits, and targeted multi-seed and data-volume controls. It also performs the provenance corrections that the earlier point estimates did not contain: the historical 0.16 eV result is not reproduced under the grouped multi-seed audit, and the historical 0.41 eV record is excluded because it is a different system with calculator and serialization defects. These additions support a narrower claim: migration pathways make model selection task-conditioned and auditable, while higher-distortion extrapolation and valid same-system self-relaxed NEB remain unresolved where the required DFT or convergence gates are not closed.

Figure/table insert and evidence boundary. No new A7-specific figure is required. The task-conditioned framework figure, fixed-path barrier tables, foundation-checkpoint profiles, and audited self-relaxed provenance panel are retained; the new Introduction paragraph and the Sec. 3.2 discussion provide the literature comparison. The nearest-neighbor evidence is comparative and contextual, not a claim that the present benchmark reproduces the protocols or materials of those studies. The manuscript therefore does not make a universal ranking claim, and the higher-distortion results remain explicitly provisional/descriptive rather than a completed extrapolation ranking.

Manuscript change.

Unchanged context: the related-work comparison is scope-setting context and does not broaden the completed evidence beyond the audited fixed-path and grouped-split results.

Inserted/revised manuscript text.

The Introduction, Sec. 3.2, softening discussion, and bibliography are marked with `REV-A7`; the former literature placeholder has been removed and the citation keys have been verified in `sn-bibliography.bib`. The related-work paragraph and the delta from the authors' earlier report are written as scope-setting context, while the quantitative conclusions remain tied to the completed fixed-path and grouped-split evidence.

Zero-force NEB exporters

Reviewer comment.

> *"Fake zero forces in NEB→training-data exporters (`examples/Sb2Te3_Cr_doped/03_neb/neb_data_colle`*
> *`neb_to_extended_xyz.py:278-282`): real QE energies paired with fabricated all-zero forces written into training-format extxyz under `forces:R:3`; identical bug in two scripts. If any such frames fed fine-tuning (`forces_weight=100` per repo README), the model was silently trained toward zero forces. Not proven wired into FT-600K/FT-MultiT (merge config references nonexistent folders). → Export real forces from QE output or hard-disable; authors must confirm/deny use in reported runs."*

Response. We agree that this was a serious code issue. We restored the two affected exporters and made them fail closed: the collector now terminates when real QE forces are unavailable, while the extended-XYZ exporter defaults to omitting forces and raises an explicit error if force export is requested without validated force data. No fabricated all-zero force array is written by either script.

We then performed a hash-bound, frame-level provenance audit of the datasets used for the reported fine-tuned models. The audit covered 9,163 frames across the historical FT-600K, historical FT-Multi$_T$, grouped FT-600K, and grouped Scratch-5% train/validation/test exports. Every frame contains a `forces:R:3` field. Exactly nine frames have forces that are identically zero; all nine are explicitly labelled one-atom `IsolatedAtom` Cr/Sb/Te reference records used for energy-offset (E_0) bookkeeping. There are zero unclassified all-zero-force frames and zero NEB-derived all-zero-force frames. Thus, the audit found no evidence that the fabricated NEB exports were included in the reported training datasets.

The release contains the complete CSV/SQLite manifest and machine-readable summary, including source-file checksums, export-time metadata, split assignments, and the acceptance result. The corresponding code and audit artifacts are in release commit `f5c6d19` (full commit: `f5c6d195a448cc1c9d5b7f5177757dc937d28276`).

Manuscript/code change.

Unchanged context: this code-provenance correction is supported by the frame-level manifest rather than by a new scientific performance figure.

Inserted/revised manuscript text.

Methods data provenance and Code Availability now describe the fail-closed exporter and the frame-level audit. No new figure is needed: the compact manifest summary and reproducibility files provide the direct evidence for this code-provenance issue.

Matched MD protocol for FT-600K

Reviewer comment.

> *"Asymmetric MD protocol for the naive-FT condition (`results/MD/2-naive-fine-tuning/mace_md_eval_` `steps=2000`, `interval=5` vs. `steps=100000`, `interval=200` in the other three byte-identical scripts): 50× shorter trajectory with 40× denser sampling in the cross-model transport comparison; post-processing assumes 200 fs/frame, so D/κ for that condition may be off ∼40×."*

Response. We agree. The original FT-600K transport value is not retained as a final model comparison because it was obtained with a shorter trajectory and a different sampling protocol from the comparison runs. In the revised manuscript, "excluded from ranking" means that the value remains in the provenance record but is not used to order models or support a quantitative transport claim.

The current local transport re-analysis is summarized below:

Table 7: Current status of transport estimates.

Model	D ($\mathrm{cm^2\,s^{-1}}$)	κ ($\mathrm{W\,m^{-1}\,K^{-1}}$)	Interpretation
Foundation	$(0.7 \pm 7.9) \times 10^{-9}$	$(2.4 \pm 8.6) \times 10^{-3}$	Statistically indistinguishable from zero within current uncertainty
Scratch	$(-3.4 \pm 3.2) \times 10^{-9}$	$(0.9 \pm 1.3) \times 10^{-2}$	Statistically indistinguishable from zero; the negative fitted estimate reflects finite-sampling uncertainty rather than a physical negative diffusivity
FT-MultiT	$(4.8 \pm 4.8) \times 10^{-8}$	$(2.0 \pm 1.7) \times 10^{-2}$	Preliminary estimate under the matched local re-analysis
FT-600K	$(3.9 \pm 4.0) \times 10^{-8}$	$(0.9 \pm 1.2) \times 10^{-2}$	Previous protocol non-comparable; provenance only, excluded from ranking

The matched-protocol, multi-seed FT-600K closure calculation remains pending. Until it is complete, the available transport curves are described as best estimates under the available trajectories, not as converged coefficients, and no model-ranking conclusion is drawn.

Manuscript/code change.

Unchanged context: the matched-protocol, multi-seed FT-600K closure remains pending, so the transport rows are not used for a model ranking.

Inserted/revised manuscript text.

The MD protocol description, transport figure, and SI transport table now identify the legacy run as provenance-only and separate it from the matched-protocol evidence.

Trajectory-grouped train/test split

Reviewer comment.

> *"Ungrouped random train/test split (`split_data.py:52-66`): frame-level shuffle over merged trajectories with no grouping by pathway/trajectory ID; near-duplicate adjacent frames can land in both train and test, inflating Table S1 test RMSE. Provenance metadata embedded by the collector (`path_name`, `neb_image`) is never read downstream—no NEB-exclusion safeguard exists."*

Response. We agree. The split has been revised to group structures by trajectory provenance, with explicit zero-overlap checks across training, validation, and test partitions, preventing information leakage through temporally or structurally adjacent configurations.

The grouped-split retraining and evaluation are complete:

Test RMSE (energy / force, meV/atom / meV/Å), mean ± std over three independent training runs: FT-600K (OMAT-FT start) 40.7 ± 5.0 / 259.7 ± 5.8; Scratch 30.5 ± 3.6 / 343.6 ± 35.9; Scratch-5% 16.7 ± 9.2 / 341.5 ± 15.0; MACE-MPA-0 comparator (single checkpoint) 38.7 / 278.5.

These grouped-split RMSE values are reported alongside the historical frame-level split values in Table S1, which is now labelled to distinguish the two splits; quantitative generalization claims in Sec. 3.3.3 use the grouped-split values.

Manuscript/code change.

Unchanged context: the grouped split prevents trajectory-adjacent leakage for the declared files, while historical frame-level values remain labelled as such.

Inserted/revised manuscript text.

The Methods split description and Table S1 now distinguish the trajectory-grouped evaluation from the historical frame-level result.

Silhouette scores in the original descriptor space

Reviewer comment.

> *"Silhouette scores computed on the 2D t-SNE embedding, not the original feature space (`results/t-SNE/3d/tsne.py:323`, `phate-tsne/tsne.py:191`, `mace_probe_blackbox0818.py:254`): the Fig. 5d separability numbers measure projection artifacts, not true latent separability. Need to compute in original/high-dimensional space."*

Response. We agree. The cited legacy scripts computed silhouette scores on the two-dimensional embeddings, so those values measure projection geometry and are retained only as projection-space baselines. The revised analysis recomputes the principal metric in the original 6,191-dimensional descriptor space, giving an Euclidean silhouette score of 0.33. The SI additionally reports cosine, z-scored Euclidean, and PCA-reduced analyses to make the dependence on feature representation explicit. The processed metrics and `docs/audit/revision1/latent_robustness_note.md` are retained in the canonical release; the legacy projection scripts are not used for the revision-facing result.

No embedding-space value is presented as true latent separability, and no force-sensitivity placeholder is included in the revised feature matrix. The remaining limitation is the absence of a new physically defined force-sensitivity calculation, not a manuscript-integration gap.

Manuscript change.

Unchanged context: projection-space values remain descriptive baselines and are not presented as true latent separability.

Inserted/revised manuscript text.

Sec. 3.4 and the Fig. 5 caption now identify the original-space metric as the primary separation diagnostic.

Inactive force-sensitivity feature stub

Reviewer comment.

> *"Always-zero 'force sensitivity' feature stub in all three latent-probe scripts (`3d/tsne.py:209-215`, `phate-tsne/tsne.py:132`, `mace_probe_blackbox0818.py:171-180`), silently concatenated into the feature matrix feeding PCA→t-SNE→PHATE."*

Response. We confirm this issue. The legacy implementation used an identity/zero-vector placeholder rather than a force-sensitivity calculation, and that channel was silently concatenated into the historical projection matrix. We removed it from the revision-facing feature matrix and quarantined the affected legacy scripts from the canonical reproduction release. The code-lineage boundary and the retained 6,191-dimensional outputs are recorded in `docs/audit/revision1/latent_robustness_note.md`. We do not make any force-sensitivity claim based on this channel, and we will not introduce such a claim unless it is supported by a genuine finite-difference or otherwise physically defined calculation.

No numerical placeholder remains in the revised latent analysis. A genuine finite-difference force-sensitivity test, including the isolated Cr–Cr perturbation requested under A6, remains a validation gap.

Manuscript/code change.

Unchanged context: no physical force-sensitivity claim is made from the historical zero-vector placeholder, and the genuine isolated Cr–Cr perturbation remains a validation gap.

Inserted/revised manuscript text.

The SI latent-analysis methods and code-release notes now remove the inactive feature from the revision-facing matrix.

SHAP code and repository completeness

Reviewer comment.

> *"SHAP analysis code absent from the repository entirely; advertised `migrationbench` package (36 files), `scripts/` (13 files including `shap_analysis.py`), templates, workflow docs, and `paper/manuscript.tex` are all 0-byte stubs. Section 3.5 results and the 'benchmark framework' API are not reproducible from this release."*

Response. The quoted zero-byte finding is correct for the pre-reproduction snapshot audited in this review, but it is not the state of the current revision release. The earlier placeholder tree was removed during the repository migration; the current release contains the X-FORCE three-model, 150-frame inputs, an explicit image-aligned SOAP/structural feature workflow, the TreeSHAP analysis scripts, pinned dependencies, and the empty-stub audit. In the current tree,

`scripts/shap_analysis.py` is a populated 5,455-byte script, `scripts/plot_shap_fig6.py` is populated, and the corresponding release commit `a3a73dfe88aa96416a3bc365909fa8165d42a271` is an ancestor of the current manuscript worktree. An explicit-input local smoke run completed and wrote the aligned SHAP outputs. The source lineage is recorded in `docs/audit/shap_source_lineage.md`; the four-model, 2-concentration, and Dual-X ID/OOD/NEB workflows are explicitly excluded from the Sec. 3.5 claim.

The locked revised CSVs report the audited cross-validated values used in Sec. 3.5. An independent local rerun reproduces the expected 150-frame/450-row alignment but is not numerically identical to those locked CSVs; this discrepancy is recorded in the lineage audit and the paper outputs are not silently overwritten pending recovery of the exact historical regeneration implementation.

The remaining release-level evidence is:

`C13 GitHub release commit: a3a73dfe88aa96416a3bc365909fa8165d42a271`

`docs/audit/c13_empty_stub_audit.md; docs/audit/shap_source_lineage.md`

Manuscript/code change.

Unchanged context: the locked SHAP outputs remain tied to their source-lineage audit, and the independent local rerun discrepancy is not silently overwritten.

Inserted/revised manuscript text.

Sec. 3.5 and Code Availability now identify the populated SHAP workflow, its source lineage, and its remaining reproducibility boundary.

References

[1] Nada Alghamdi, Paolo de Angelis, Pietro Asinari, and Eliodoro Chiavazzo. Comparing fine-tuning strategies of mace machine learning force field for modeling li-ion diffusion in lif for batteries. *arXiv preprint arXiv:2510.05020*, 2025.

[2] Achinthya Krishna Bheemaguli, Penghao Xiao, and Gopalakrishnan Sai Gautam. Evaluation of foundational machine learned interatomic potentials for migration barrier predictions. *Digital Discovery*, 5:1809–1819, 2026. doi: 10.1039/D5DD00534E.

[3] Bowen Deng, Yunyeong Choi, Peichen Zhong, Janosh Riebesell, Shashwat Anand, Zhuohan Li, KyuJung Jun, Kristin A. Persson, and Gerbrand Ceder. Systematic softening in universal machine learning interatomic potentials. *npj Computational Materials*, 11(1):9, 2025. doi: 10.1038/s41524-024-01500-6.

[4] Yi Cao and Paulette Clancy. Migration as a probe: A generalizable benchmark framework for specialist vs. generalist machine-learned force fields in doped materials. *arXiv preprint arXiv:2509.00090*, 2025. URL https://arxiv.org/abs/2509.00090. AI for Accelerated Materials Discovery (AI4Mat) Workshop, NeurIPS 2025; non-archival workshop report.